

TPE@LU ASTM F2291 Compliance

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SCOPE

This document summarizes the design, manufacturing, and testing of Joop Adventures. This fully functional 3D-printed roller coaster will transport Jolly Rancher Chews in a Joop ride vehicle on a wild jungle adventure. Riders will begin by being lifted 417 mm above the station and 445 mm above the baseplate on a 90-degree lift hill, then go around a downward helix, achieving at least 2G of absolute acceleration. As they leave the helix, they will enter a 45-degree outer bank, then a magnetic braking section before re-entering the station. The ride will remain fully functional within the required 750 x 750 x 900 mm operational envelope and can be disassembled and placed inside a portable envelope measuring 700 x 250 x 250 mm.

All design criteria will follow the ASTM F2291 standards, unless explicitly stated to be omitted in the Ride Engineering Competition (REC) 2026 ASTM Adaptation Document. This ride is designed to operate safely within the specified spacing constraints.

REFERENCED DOCUMENTS

- [1] ASCE 7-22
- [2] ASTM F2291
- [3] ASTM F1193
- [4] ASTM 0771
- [5] REC Ride prompt
- [6] REC ASTM Compliance Adaptation

TERMINOLOGY

All terminology in this document applies to candied riders, not actual patrons. Areas such as the rider envelope and clearance area will be addressed in accordance with the

dimensions provided in the REC 2026 ASTM Adaptation Document.

All lengths are given in millimeters, acceleration in G, and angles in degrees, unless otherwise stated.

SIGNIFICANCE AND USE

The purpose of this document is to summarize the design, manufacturing, and testing of Joop Adventures to present a fully functional ride for the Ride Engineering Competition 2026. This document is intended for review by the REC Planning Committee, which will verify that Joop Adventures complies with ASTM and competition standards.

GENERAL DESIGN CRITERIA

All terminology used in this document will apply to our candied riders, not to actual patrons. Areas such as the rider envelope and clearance area will be addressed in accordance with the dimensions provided in the REC 2026 ASTM Adaptation Document.

5.1 Ride Analysis: The ride analysis for Joop Adventures demonstrates how hazards to the riders are detected and minimized.

5.1.1 The engineers shall prepare a ride analysis summary that outlines how potential risks have been managed and mitigated. (See Figures 4-10 in the Appendix.)

5.1.1.1 The scope of analysis includes all components that pose a hazard to rider safety, including the electrical, mechanical, and structural components within the Joop Adventures assembly.

5.1.1.2 Identification of hazards, including potential sources and consequences of harm can be found in the Appendix Figures 4-10 below.

5.1.1.3 Hazardous scenarios comply with the requirements listed in the ASTM F22291 document, including:

(1) Hazardous scenarios shall consider physical areas where persons are expected to access.

(2) Hazardous scenarios shall consider exposure to hazards under all identified use cases.

(3) Hazardous scenarios related to maintenance personnel actions shall be restricted to routine, repetitive and essential procedures.

The complete hazard identification and mitigation can be viewed in the Appendix (Figures 4-11).

5.1.1.4 Hazards identified are mitigated to an acceptable level to reduce the risk of injury. See Appendix (Figures 4-10) below.

5.2 Hazard Mitigations shall consider:

5.2.1.1 The primary lap restraint system experiences a maximum stress of 8.05 MPa, alternating stress of 4.03 MPa, and mean stress of 4.03 MPa. The calculated fatigue factor of safety is 2.87, the predicted cycles to failure are 5.84×10^9 and the required cycles are 2.25×10^5 based the full lifetime of 224640 cycles (3 years and 2 hours a week). The result was that the design for the lap restraint system meets the lifetime requirement. These values were determined from the complete track analysis MATLAB document where all the normal and lateral forces were tabulated. See the MATLAB file for additional information.

5.2.1.2 Due to environmental conditions: A fatigue analysis was performed on the primary lap restraint using the Goodman mean stress criterion and a Basquin high-cycle fatigue model. The calculated maximum stress in the component was significantly below the ultimate tensile strength of PLA, resulting in a high static factor of safety under nominal material conditions.

Under nominal conditions, the predicted fatigue life exceeds the required lifetime loading of 224,640 cycles. The Goodman fatigue safety factor is greater than unity, indicating that the alternating and mean stresses remain within acceptable fatigue limits for high-cycle operation.

To comply with ASTM design intent for amusement ride structures (particularly ASTM F2291), environmental and material variability factors were incorporated into a separate reduction analysis. Although the ride will be operated indoors, the following conservative reductions were applied:

Indoor temperature variation

Indoor humidity exposure

FDM print anisotropy

Long-term creep effects in PLA

Manufacturing variability

These reductions were combined into a single material knockdown factor, reducing the effective ultimate and fatigue strengths of the material.

After applying these reductions:

The static factor of safety remains above acceptable structural limits.

The fatigue factor of safety remains greater than 1.0.

The predicted fatigue life continues to exceed the full design lifetime of 224,640 cycles.

This demonstrates that the restraint maintains adequate structural capacity and fatigue resistance even under conservative indoor environmental and material degradation assumptions.

Because the competition cycle requirement (4,320 cycles) is significantly lower than the lifetime requirement, the design inherently satisfies the competition's durability requirement as well.

Overall, the analysis confirms that the 3D printed PLA lap restraint meets structural and fatigue performance expectations for indoor operation when evaluated under conservative ASTM-aligned reduction factors. The combined reduction factor was 0.49, static FOS (reduced) was 0.49, fatigue FOS (reduced): 1.41, and predicted life (reduced): 1.98×10^7 cycles.

5.2.1.3 The IR break beam sensor used for the ride control system was analyzed for mechanical safety under sustained loading conditions in accordance with ASTM F2291 recommendations. The sensor is mounted to a copper support column using a 3D-printed PLA bracket. The maximum sustained force transmitted through the bracket to the sensor is 1.04 N.

A conservative allowable static load for the sensor casing was assumed to be 5 N, consistent with typical small infrared sensor specifications. The corresponding mechanical factor of safety is calculated is 4.8.

The expected ride lifetime is 224,640 cycles, which is well below the endurance limits of the sensor under the applied load. Therefore, the sensor is expected to maintain reliable mechanical and functional integrity throughout the ride's operational life.

The IR break beam sensor, as mounted on the PLA bracket, satisfies mechanical safety requirements under ASTM F2291 for the intended indoor roller coaster application. The factor of safety of 4.8 provides a substantial margin against mechanical failure for the full ride lifetime. Refer to the MATLAB code for detailed calculations and results.

5.2.3 Hazard mitigations that involve operator responsibilities:

5.2.3.1 Operator actions and responsibilities are outlined in the appendix in figures 4-10.

5.2.3.2 Operator responsibilities related to hazard mitigation are outlined in the appendix in figures 4-10.

5.2.4 Hazard mitigations that involve operator procedures or involve routine, repetitive, or essential maintenance procedures shall:

5.2.4.1 Identify the frequency of procedures (see Appendix Figures 4-10)

5.2.4.2 Defined elements that must be included in documented procedures to reduce associated hazard (see Appendix Figures 4-10)

5.2.5 The ride is equipped with control devices that allow the operator to safely start, stop, or interrupt ride operation at any time. These devices are located at the designated operator control station, which provides a clear line of sight to all areas of the ride and the train during operation.

The control system includes:

Emergency stop button at the operator station

IR break beam sensors along the track to detect train position and prevent unsafe dispatch

PLC logic to automatically halt the ride if a train is detected outside safe zones

The operator's responsibilities, including monitoring train position, verifying restraint engagement, and initiating emergency stops, are supported by both visual and sensor feedback. The operator is positioned within the identified zone of awareness, ensuring that intervention is possible immediately when needed.

This design meets the requirements of ASTM F2291 section 5.2.5 by providing appropriate devices and controls for operator intervention while considering operator location and visibility of the ride.

5.3 Design and Calculations:

5.3.1 Calculations were performed to show compliance with the latest version of ASTM F2291 document. (See MATLAB and Excel Documents for further information).

5.3.1.1 All structural support columns and mechanical components of the track have an adequate factor of safety and can sustain the competition duration of 6 hours as well as comply with a lifetime of 3 years and 2 hours a week. See MATLAB and Excel files for detailed calculations.

5.3.1.2 Acceleration calculations were done in Excel and reference the data in MATLAB for G forces experienced on the ride.

5.3.1.3 Joop Adventures is equipped with safety break beam sensors and procedures that allow the operator to safely enable, monitor, and, if necessary, interrupt ride operation.

Before each ride cycle, the operator manually verifies that all restraints are properly engaged. The lap restraint incorporates a ratcheting locking mechanism to secure the waist, while the over-the-shoulder restraint uses a latching design to ensure the upper body is held safely. This manual verification procedure ensures that every passenger is properly restrained prior to ride dispatch.

The operator control station is positioned to provide clear visibility of the train and restraint areas, allowing timely intervention if unsafe conditions are observed. The control system includes IR break beam sensors along the track to detect train presence, verify proper train spacing, and prevent ride

dispatch until all conditions are safe. In addition, the system includes an emergency stop function that allows the operator to immediately halt ride operation if a hazard is detected.

Critical control system functions, including train detection, restraint verification, and emergency stopping, are implemented with fail-safe logic and redundancy to maintain functionality even if a single component fails. Operational feedback, including visual indicators at the control station, assists the operator in confirming the status of restraints and ride readiness.

5.3.1.4 In accordance with ASTM F2137, all structural and fatigue calculations for the ride system were performed using the x-, y-, and z-axes established by the Ride Engineering Committee (see Figure 1 for axes and MATLAB and Excel Documents for further information).

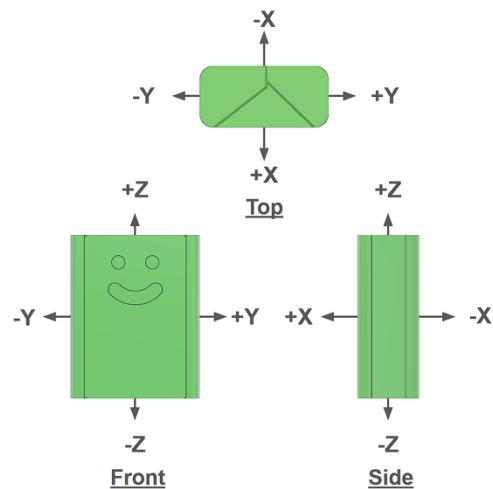


Figure 1: Coordinate system used as the standard reference for acceleration directions, adapted from the Ride Engineering Competition (REC 2026) ASTM Adaptation Document. The axes (X, Y, Z) define the longitudinal, lateral, and vertical directions relative to the rider.

5.4 Omitted as Joop Adventures does not encounter water.

5.5 Units:

5.5.1 All units, unless otherwise specified, are in metric (mm, MPa, kg, etc.)

5.5.2 The coordinate system shown above in Figure 1 shall be used as the standard reference for acceleration directions.

5.6 All drawings shown below utilize metric system standard units.

5.6.1 See SolidWorks Drawing package for ASME Y14.100 drawings. These requirements pertaining to retaining the drawings for 20 years do not apply to our ride. The REC 2026 ASTM Adaptation Document states that this section does not fall within the scope of the Ride Engineering Competition.

5.6.2 This requirement does not apply to our ride. The REC 2026 ASTM Adaptation Document states that Section 5.6.2 does not fall within the scope of the Ride Engineering Competition.

5.6.3 SolidWorks drawings are complete with dimensions, appropriate notes, and adequate information regarding the material. (See SolidWorks Drawing package for more information).

5.6.4 Appropriate dimensions are supplied in the SolidWorks drawings compliant with the ASME Y14.100 standard.

5.6.4.1 The complete SolidWorks assembly drawings are included in the SolidWorks drawings package.

5.6.4.2 Sub assembly drawings are provided for additional clarification within the SolidWorks drawing package.

5.6.4.3 All components including manufactured and over the counter components have drawings within the SolidWorks drawing package.

5.7 Regulated Body Review Documents:

5.7.1 The following documents are made available for review:

5.7.1.1 General assembly drawings are found within the SolidWorks Drawing Package.

5.7.1.2 Facility interface drawings are available within the SolidWorks Drawing Package and the related load calculations are in the MATLAB and Excel documents.

5.7.1.3 Operations Instructions

The ride is operated by a single trained operator who is responsible for the safe loading, dispatching, and monitoring of all ride vehicles. Each ride cycle begins with passengers being seated and properly secured. The operator manually verifies that all passengers' primary ratcheting lap restraints are locked and that the over-the-shoulder latching harnesses are properly engaged. Ride vehicles are held stationary in the station by friction wheels until dispatch, which is controlled by servo motors operating the station gates. A servo motor then lifts the ride chain up the 90-degree lift hill to begin the ride. The track features a 45-degree banked turn, a tapered helix around a volcano with integrated fog effects, and an MP3 audio system. A magnetic braking section in the laboratory ensures controlled deceleration. IR break beam sensors along the track monitor vehicle positions, preventing unsafe dispatch of the second ride vehicle and enabling automatic ride halting if necessary. An emergency stop button located at the operator station allows immediate interruption of ride operation. The operator maintains continuous line-of-sight to the ride vehicles and track zones to ensure safe operation.

Maintenance Instructions

Before each ride cycle, the operator inspects all 3D-printed lap restraints for cracks, wear, or deformation and ensures the over-the-shoulder latching harnesses operate correctly. IR break beam sensors and their 3D-printed brackets are checked for damage, alignment, and secure attachment to support columns. Friction wheels, servo motors controlling the station gates and lift hill, and the magnetic braking section are verified for proper operation. The emergency stop button is tested to ensure immediate response. Periodically, PLA components are cleaned

with a cloth, servo motors and electrical connections are inspected for wear or looseness.

Audio and fog systems are additionally monitored to ensure these systems are operating appropriately. All fasteners securing restraints, sensors, and structural supports are tightened as necessary. Replacement parts, including spare lap restraints, harness components, and sensor brackets, are kept on hand, and any worn or damaged parts are replaced immediately following inspection.

Chain link will be inspected to ensure that the chain is not rusting or cracking. If either one of these issues are present the ride will cease operation until the chain can be cleaned or replaced. Additionally, lubrication checks will be conducted periodically to ensure the chain is operating appropriately.

Assembly Instructions

Ride vehicle assembly begins by attaching the lap restraints to each vehicle seat and confirming that the ratcheting mechanisms function smoothly. Over-the-shoulder harnesses are installed and latched securely. Vehicles are aligned on the track so that friction wheels in the station correctly hold the cars. IR break beam sensors are mounted on 3D-printed brackets and attached to support columns at designated detection zones, with alignment verified for proper beam activation. Servo motors controlling station gates and the lift hill chain mechanism are installed, and the MP3 audio system and fog machine are connected and secured. A full ride cycle is tested manually to confirm correct operation of all restraints, sensors, gates, braking systems, and emergency stops. The operator's line-of-sight to all ride zones is verified to ensure immediate intervention capability.

5.7.1.4 This requirement does not apply to our ride. The REC 2026 ASTM Adaptation Document states that Section 5.6.2 does not fall within the scope of the Ride Engineering Competition.

5.7.2 Use of manufacturer's documentation and records shall be limited to minimize disclosure of proprietary and confidential documents.

PATRON RESTRAINT, CLEARANCE ENVELOPE, AND CONTAINMENT DESIGN CRITERIA

6.1 Patron Containment:

6.1.1 Patron containment systems are essential to rider safety as these devices lock or latch the rider securely in the ride vehicle to prevent ejection and to transport them throughout the duration of the ride safely. The containment systems used on this ride are based on the ranges shown in Figure 2 below. Our ride is classified under the class 4 restraint system (according to the MATLAB calculations), however our team decided to manufacture a class 5 restraint system to better protect the riders and challenge our manufacturing abilities.

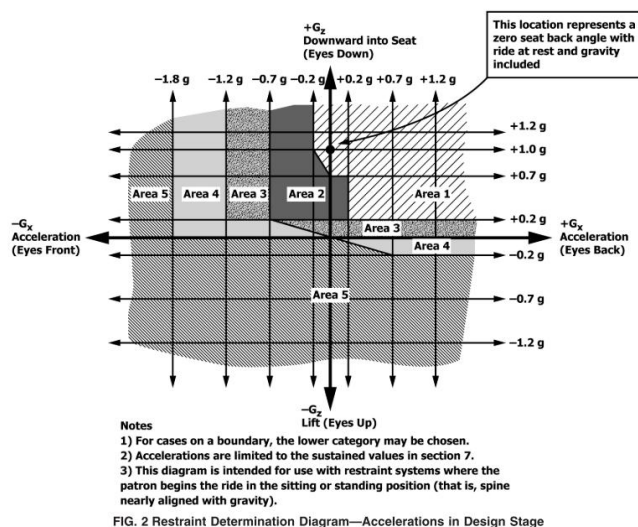


Figure 2: Restraint diagram showing over-the-shoulder and waist restraint placement for the Jolly Rancher Chew surrogate model, adapted from ASTM F2291 and the Ride Engineering Competition (REC 2026) ASTM Adaptation Document. Restraints are positioned to ensure containment and alignment with the standard's criteria for acceleration directions and patron safety. With our current G forces, our ride falls into the class 4 restraint system, but we have opted for a class 5 restraint mechanism to challenge our team and better protect the riders.

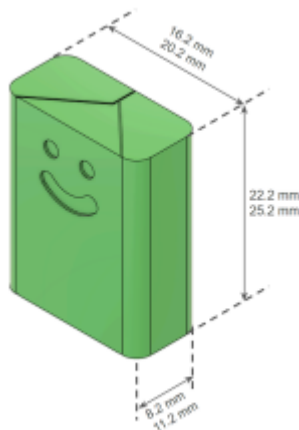


Figure 3: Rider Dimensions - taken from REC 2026 Prompt

6.1.2 For Joop Adventures, all components where riders will come into contact, such as loading, unloading, and evacuation, have been designed to comply with section 6.1.2.

Areas where patrons will meet an element of the ride include, but are not limited to, ride vehicle seating surfaces, Interior vehicle panels, lap bar restraints, loading platform edges, stairs, railings, and the baseplate with outlined paths leading to and from the station.

Fasteners located within reach are covered with protective caps. Edges of sharp materials are sanded to prevent injury. Materials selected for this ride are resistant to splintering and degradation.

Residual risk is considered low due to the mitigation in place to prevent injury.

6.1.3 Omitted as Joop Adventures has an adequate patron carrier and ride vehicle.

6.1.4 Ride or Device Vehicle Doors:

6.1.4.1 Omitted as Joop Adventures does not have ride doors.

6.1.4.2 Omitted as Joop Adventures does not have ride doors.

6.2 Security of Patron Containment Systems:

6.2.1 Ride seats, restraints, and seating structures shall be securely attached to the frame of the ride vehicle.

6.3 Patron Restraints

6.3.1 Rider characteristics shown in Figure 1 shall be used to determine the appropriate seat restraint.

6.3.2 The analysis of the seat restraint considered seat inclination, ride orientation, emergency stop, and G forces experienced throughout the duration of the ride, as shown in the MATLAB and Excel calculations as well as the MATLAB and ANSYS calculations (Appendix Figures 11-32). The results displayed a class 4 restraint system, but (as previously stated 6.1.1), the Class 5 restraint system was chosen for enhanced safety and to present more of an engineering challenge. A bottom ratcheting locking lap bar will secure the rider's waist, sidewalls will prevent lateral motion, and a top latching lap bar will serve as a secondary method for securing the rider in their seat.

6.3.3 Maximum allowable restraint travel is 2 mm. Lap restraints are visually verified to be securely in place before the ride vehicle exits the station.

6.3.4 In the event of an emergency, the bottom lap bar remains locked, and the top lap bar remains latched. An authorized operator will remove these restraints.

6.3.5 Restraint mechanisms have been designed to mitigate pinching and unintentional trapping of appendages.

6.3.6 Omitted as Joop Adventures does not utilize a powered patron restraint device.

6.3.7 Load calculations related to the lap restraint are in the MATLAB code.

6.3.8 Both the primary and secondary restraint harnesses are released manually by the operator.

6.3.8.1 The manual release is conveniently located for the operator without being able to be released by the patron. The release mechanism does not require the operator to crawl or come into direct contact with the patrons to release the restraint.

6.3.8.2 External or unmonitored internal nonmechanical stored energy will not be used for manual release on Joop Adventures.

6.3.8.3 Special tools are not required to operate the manual release.

6.3.9 According to our patron restraint and containment analysis, a gap of 5 mm will be between each seat side to prevent lap bar mechanisms from colliding. Additionally, sidewall containments will be 3D printed on the frame of the ride vehicle to better secure the rider from any lateral forces experienced.

6.4 Restraint Configuration

6.4.1-6.4.2.2 Omitted as Joop Adventures is not considered a kiddie ride due to the G forces experienced (see MATLAB and Excel calculations).

6.4.3 Restraints Required Due to Acceleration and Seat Inclination:

6.4.3.1 Class 5 restraint systems are utilized to prevent rider ejections including locking and latching lap bars.

6.4.3.2 The restraint diagram in Figure 2 and the MATLAB code were used to determine the appropriate class of restraint harnesses for Joop Adventures.

6.4.3.3-6.4.3.7 These sections do not apply to Joop Adventures, as it will use a Class 5 restraint system.

6.4.3.8 Due to the MATLAB and Excel calculations determined that Joop Adventures will operate with a class 5 restraint system.

- (1) Each rider will have an individual restraint system.
- (2) The restraint system for the bottom locking waist restraint system will ratchet into place, securing the riders.
- (3) Joop Adventures is unable to incorporate an automatic locking restraint system due to the scope of the competition. Implementation of an automatic locking restraint system would require an onboard computer small enough to fit within the ride vehicle. It would be cost-effective and ensure it does not exceed budget limits. A manually activated restraint system was selected to provide adequate patron containment.
- (4) Both the top and bottom lap restraint systems will be unlocked and locked by an operator.
- (5) Visual checks will be performed by the operator prior to the ride vehicle exiting the station.
- (6) The restraint will be manually opened and closed.
- (7) Locking and unlocking of the secondary restraint system is independent of the primary restraint system. The secondary restraint system latched into a groove in the side of the seat, separate from the primary restraint system.

The lap restraint system can be seen in the Appendix in (*Appendix Figures 33-34*).

6.4.3.9 Secondary Restraints for Class 5-A

The secondary restraint configuration to complete our two independent restraint system is a U-shaped over-the-shoulder restraint. While this is not required by the Ride Engineering Competition, the secondary lap restraint was included to increase the safety of the ride vehicle system and challenge the students on the team to manufacture a safer ride.

- (1) Number of Patrons per Restraint Device – Each restraint device will be assigned to each patron.
- (2) Final Latching Position Relative to the Patron- The final latching position will be fixed to a clamp proportioned to the maximum depth of the patron, as seen in Figure 18.
- (3) Type of Locking- The restraint device will be manually locked into place by a member of ride operations
- (4) Type of Unlocking- Restraint devices shall be unlocked manually by a member of ride operations.
- (5) Type of External Correct or Incorrect Indications – No external indication will be required for the secondary restraint device other than a member of the ride operations team visually checking that the over-the-shoulder restraint is appropriately attached to the clamp.
- (6) Means of Activation- The restraint will be manually opened and closed by a member of ride operations.
- (7) Redundancy of Lock Device- The over-shoulder restraint will not require any redundancies, as the locking and unlocking of the device will be independent of the primary restraint.

6.5 All physical information provided with practice to the ASTM F1193 compliance documentation shall be consistent with all patron restraint systems.

6.6 Patron Clearance Envelope Analysis

6.6.1 The patron clearance envelope, according to the Ride Engineering Competition Prompt and Adaptation Document, was considered during the design and manufacturing of Joop Adventures. This includes a 10 mm clearance around the ride vehicle and an additional 1.5 mm clearance to include an additional layer of safety.

6.6.2 When the patron encounters the ride vehicle or station, the team ensured that the station was sanded and chamfered, the station stairs are not at a sharp angle and feature non-sharp points of contact, the ride vehicle contact points do not include any sharp points of contact, and the baseplate featured clearly marked entry and exit paths that were sanded and painted to prevent tripping and falling risks.

6.6.2.1 Patron-to-patron contact does not occur in Joop Adventures, the gap between ride seats is 5 mm and the patron reach is only 1.5 mm.

6.6.3 To ensure the quality of our ride experience, we have dedicated time to finding the optimal spacing required to limit patron-to-patron contact within the given scope of documents [4],[5], and [6]. According to these documents, the rider should have a minimum of 11.5 mm clearance, which was considered when designing and manufacturing Joop Adventures.

6.6.3.1 In document [1], the dimensions of the patrons shown in Figure 2 are given, with patron weights ranging from 4 to 5 grams. Additionally, information in document [2] indicates that all patrons have a relative arm length of 10mm, and that an additional 1.5 mm from the end of the arm was required to create a patron clearance envelope. For this reason,

patrons who do not meet the dimensions listed in documents [1] and [2] cannot ride this ride.

6.6.3.2 All information regarding shapes and configuration of the ride vehicle can be found in sections 6.3 to 6.4 of this document.

6.6.3.3 All attributes that the ride that encounter the ride including the station, stairs, ride vehicle, and baseplate have been appropriately accounted for stated in section 6.6.2

6.6.3.4 Appropriate calculations were determined to ensure that the ride vehicle would not enter the station at high speeds which would increase the risk that they would contact a part of the station at high speeds.

6.6.3.5 Infrared break beam sensors are positioned at key places along the track, including at the beginning and end of the lift hill, before the helix, and before the magnetic braking section to ensure that the two ride vehicles do not collide with each other along the track and remain within the appropriate time limits. Additionally, scenic elements that are located along the baseplate are placed so that they are located outside of the ride vehicle clearance envelope and will not contact the rider or ride vehicles.

6.6.3.6 The primary ratcheting restraint mechanism features multiple levels where the restraint harness can lock into place to ensure that riders within the given dimensions in the prompt [1] are secured fastened into the ride vehicle.

6.6.4 The patron clearance envelope has been confirmed within the modeling software as well as using physical measuring devices, such as a tape measure, were utilized to ensure that the rider and the ride vehicle would not unintentionally encounter any elements of the ride.

6.6.5 To ensure that no possible physical elements can encounter the ride vehicle. The team modeled Joop Adventures in SolidWorks. From the start of the reach envelope, we ensured that structures not designated as loading/unloading zones were at 11.5 mm from the start of the patron reach envelope to prevent any possible contact per the Prompt [1] and Adaptation documents [2].

6.7 Signage – Throughout our thematic queue line, signage will be presented to show the requirements that all riders must meet to be able to ride Joop adventure, with additional restrictions to patrons with pre-existing medical issues that could put them in harms way of further injury if they plan to ride Joop Adventures, and that loose articles such as hats, glasses without straps, and phones, etc., will not be permitted on the ride as they could cause possible endangerment to the patron envelope if it were to fall out of the patrons hands or pockets.

Additional signage for ride-restricted areas will be present to prevent patrons from entering maintenance/operator-only areas that could cause injuries.

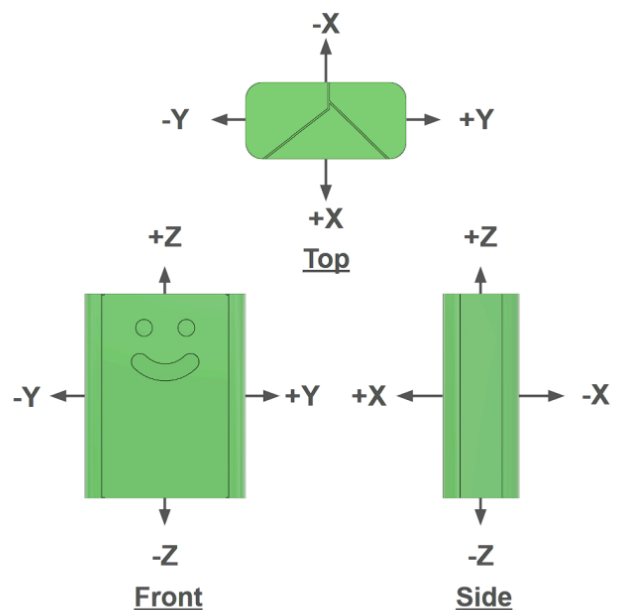
ACCELERATION LIMITS

7.1.1-7.1.1.3 – The following sections have been omitted in accordance with document [6] as the test data will

not be required. However, our team has performed extensive MATLAB and Excel calculations to quantify the accelerations experienced by the riders.

7.1.2 Joop Adventures will not operate outside ASTM's acceleration standards. Calculations can be viewed in the MATLAB and Excel files. Parts of the calculations or graphs that exceed the provided limits were classified as a jerk, which sustain acceleration less than 0.2 seconds. All the sustained portions of Joop Adventures comply with ASTM's standards.

7.1.3 – In Figure 3 below provided in document [2], we are given a coordinate system that displays the physical orientation of our rider during operation. From the information given, combined with the acceleration that we calculated, we created a patron containment system suitable to the forces and acceleration of our ride.



7.1.4 – The sustained acceleration duration limits of Joop Adventure are shown in the appendix figures 23 - 25

7.1.4.1 – Acceleration units are “g” (9.81m/s^2).

7.1.4.2 - The sustained acceleration duration limits of Joop Adventures can be seen in the appendix figures 23-25

7.1.4.3 – Omitted by REC ASTM Adaptation document [2].

7.1.4.4 – Due to insufficient information about patrons with physical limitations, the given acceleration data did not account for patrons with these limitations.

7.1.4.5 – The specified axes have been limited to the total net acceleration, inclusive of Earth's gravity, so a motionless patron will be measured to have a magnitude of 1 g measured perpendicular to the surface of our base plate and 0 g's parallel to the axis of our base plate.

7.1.4.6 – Steady-state values shown in the figures will be limited to time, and sustained forces exceeding 0.2

seconds will be addressed in the containment analysis and further explained in the risk assessment.

7.1.4.7 - The limits shown in [Insert Figures] are proportional to the class 5 restraint system created.

(1) Due to the acceleration and G force limits of Joop Adventures, a ratcheted lap bar is required to support our patrons' lower bodies in all directions and maintain contact with the rider containment vehicle.

(2) An over-shoulder restraint will also be provided to support the upper torso of our patrons due to the sustained acceleration and forces of Joop Adventures, as seen in *figure 23* in the appendix.

7.1.4.8 – A slope of best fit was made to show the total acceleration over time of Joop Adventures through a 9.7 second window. This graph can be seen in *figure 32* in the appendix.

7.1.5 – 7.1.5 - To adhere to section 7.1.5 of ASTM standards, we measured the combined acceleration of Joop Adventures on multiple axes with our base and upper torso restraint. In *figures 25-28* in the appendix, elliptical limitation charts are shown, depicting the acceleration limits for both the primary and secondary restraints with respect to the patron coordinate system: eyes right, eyes left, eyes up, and up-down.

The elliptical boundary for each quadrant is centered at (0,0) and characterized by major and minor radii equal to the allowable 200 ms g limit. Joop Adventures successfully falls within this range, not allowing for any data to be excluded from the chart.

7.1.6 – In this section, we will discuss the reversals of acceleration of the X & Y values.

7.1.6.1 – According to ASTM F2291, the peak-to-peak transition time between consecutive sustained events in the X & Y accelerations needs to be greater than 200 milliseconds. To ensure we meet this standard, we calculated in MATLAB the time the ride vehicle spends in each section. In [Insert Figure], the elapsed time for each section was calculated to determine the average transition time. Once all the data were collected, they were entered into a frequency chart, as shown in *Figure 30* in the appendix.

7.1.7 - Transitions in Z

7.1.7.1 – In accordance with section 7.1.7.1 of document [6], transitions directly from negative (eye up) limits and positive (eyes down) limits will be restricted. At any point during the Joop Adventures ride cycle, if patrons experience 3s of negative Gz, our +Gz limits will be reduced by 6s for 6s after the transition to positive transitions.

7.1.7.2 – All alternative transitions in z acceleration are shown in the appendix *Figure 31*. In accordance with ASTM F2291 standards, Joop Adventures must sustain weightless (0g) and more negative levels to 2g and more positive levels, with a minimum duration of 133ms. Figures 30 and 31 in the appendix show the values obtained and indicate that all sections with high transitions meet ASTM standards.

7.2 Exceptions

7.2.1 – Joop Adventures only exemption to this standard is that comparative to a normal human patrons or riders are candied riders so the ride containment configuration will be different from normal ones as we must accommodate different rider heights and seating coordinates.

7.2.1.1 – This section has been omitted by the REC 2026 ASTM adaptation document due to not needing a

LOADS AND STRENGTHS

8.1 Overview:

8.1.1 This section will define all the loads that must be applied, the strength criteria the materials and structures must meet, the calculation methods that must be used to verify safety, and the requirements for new designs and modifications.

8.2 General:

8.2.1 Joop Adventures has been designed to avoid operational loads from causing failures, yielding, and permanent damage to components.

8.2.1.1 Joop Adventures has been designed to avoid operational loads from causing failures, yielding, and permanent damage to components.

8.2.1.2 There are no components that are needed to yield intentionally.

8.3 6 Operational hour criteria:

8.3.1 - 8.3.2 The primary structures, such as the track and support, have been designed to last the minimum operational criterion of 6 hours. The operational hour of Joop Adventures is the total amount of time the ride is in operation, which is included in the operational hour calculations.

8.3.2.1 - 8.3.3 Please refer to the MATLAB code for the specific calculations of the operational hour criteria.

8.4 Exceptions to the 6 Operational Hours Criteria:

8.4.1 Some of the components that are excluded from the 6 operational hours requirement include wheel bogies, electrical components, and the drive tires to move and stop the vehicle in the station.

8.4.2 Some of the components that are excluded from the 6 operational hours requirement include wheel bogies, electrical components, and the drive tires to move and stop the vehicle in the station. These components are all able to be replaced, so they are excluded.

8.4.3 These components are exempt from the 6 operational hour requirement; however, they are not exempt from the strength requirements, the impact factors, the dynamic load criteria, and the environmental load criteria.

8.5.1 Omit; Out of Scope

8.6 Patron Weights:

8.6.1 The upper weight of a rider, or an adult patron, is 5 grams.

8.6.2 The upper weight of a rider, or an adult patron, is 5 grams according to the prompt [1]

8.6.3 The fatigue for everything that is designed for every rider is designed to operate with an upper load of 5 grams at every position of every available seat position.

8.6.4 The fatigue has been considered in the case of adult patrons causing the Joop Adventures car to be unbalanced.

8.6.5 The fatigue for everything that is designed for every rider is designed to operate with a lower load of 4 grams at every position of every available seat position.

8.6.6 The fatigue has been taken into account in the case of child patrons causing the Joop Adventures car to be unbalanced.

8.6.7 There are no specific limitations in the calculations of the loads.

8.6.8 Omit; Out of Scope

8.6.8.1 Omit; Out of Scope

8.7 *Loads:*

8.7.1 All loads have been considered throughout the design and manufacturing of Joop Adventures.

8.7.2 Refer to the MATLAB code for the calculations of the loads.

8.7.3 Empirical tests, such as measuring acceleration and weighing the ride vehicles, have been performed to verify the calculations and prevent purely theoretical design errors.

8.8 *Permanent Loads:*

8.8.1 The permanent loads include the weight of the ride vehicles, the weight of the structure, and the weight of the equipment, such as the chain for the lift hill.

8.9 *Variable Loads:*

8.9.1 The variable loads include operational loads, nonoperational loads, environmental loads, and the operation of the ride in wind.

8.10 *Operational (Dynamic) Loads:*

8.10.1 The operational loads include the G-forces, the acceleration forces, and the braking forces.

8.10.2 High cycle and low cycle dynamic loads have both been considered.

8.11 *Nonoperational Loads:*

8.11.1 The nonoperational loads include the transportation loads, the maintenance loads, and the setup and teardown loads.

8.12 *Environmental Loads:*

8.12.1 For a portable ride, the environmental loads that have been considered include wind, temperature, and seismic loads.

8.12.2 This is omitted since Joop Adventures is not a fixed or permanent structure.

8.12.3 The environmental loads that were used include wind, temperature, and seismic loads.

8.13 *Operation in Wind:*

8.13.1-8.13.2 Joop Adventures is designed to withstand any amount of wind that can be encountered indoors.

8.14 When the wind becomes too high of a speed, the ride will be closed until the speed lowers down to an acceptable speed for a certain amount of time.

8.15 *Design:*

8.15.1 There is a structural analysis applied to this ride. Please refer to the MATLAB code for the specific calculations.

8.15.2 The calculation that was used can be shown in MATLAB.

8.15.3 The structural analyses performed consider and incorporate every significant load, stresses, and strains stated in section 8.7.

8.15.4 A structural analysis has been applied to the plastic track as seen in the Appendix, *Figures 17-22*.

8.15.5 A deflection analysis has been performed. Please refer to the MATLAB code for the specific calculations.

8.15.6 The structural analysis considers the strength and fatigue criteria in the evaluation of stresses. Please refer to the MATLAB code for the specific calculations.

8.15.7 The yield strength, ultimate strength, and fatigue properties can all be found in MATLAB.

8.16 *Impact Factor for Strength and Fatigue Analysis:*

8.16.1 A minimum impact factor of 1.2 will be used.

8.16.2 The impact factor is a minimum value of 1.2, so this standard does not apply.

8.16.3 During testing, the calculations will be revised if the forces are higher than what was calculated.

8.16.4 There will be modifications in the structure if there are any structure deficiencies in the revised calculations. Then, testing will be repeated.

8.16.5 As components wear, the impact and vibration loads may increase. So, these loads have been considered throughout the design and manufacturing process.

8.17 *Anti-rollback devices:*

8.17.1 There will be a minimum impact factor of 2.0 applied to the anti-rollback device.

8.17.2 The fatigue of this anti-rollback device will be considered.

8.18 *Vibration Factor for Structural Ride (or Device) Track Components for Strength and Fatigue Analysis:*

8.18.1 A vibration factor of 1.2 has been applied to dynamic loads.

8.19 *Resonance Protection:*

8.19.1 There are no special additional provisions needed.

8.20 *Serviceability:*

8.20.1 Omit; Reference not Required

8.20.2 The design of the support structure and bases has been designed with adequate rigidity and stiffness to maintain the alignment of movable components.

8.21 *Design for Strength:*

8.21.1 There are two accepted methods, known as Load and Resistance Factor Design (LRFD) and Allowable Stress Design (ASD). ASD is being used for assuring adequate strength for Joop Adventures.

8.21.2 The load factors and allowables from the Allowable Stress Design (ASD) are shown in MATLAB.

8.22 *Load Combinations for Strength Using ASD:*

8.22.1 Dead load: Includes copper tubing supports, 3D printed track, and wooden station.

Live load: Includes operational loads, nonoperational loads, environmental loads, and the operation of the ride in wind.

Roof load: Includes roof to the station and lab.

Wind load: This is being considered.

Earthquake load: Earthquake testing will begin when ride testing begins.

Fluid load: This is omitted since there are no hydraulics nor any fluids being used for Joop Adventures.

Earth loads: This is omitted since Joop Adventures will be on top of a wooden base.

Self-straining loads: Includes the changes due to the temperature and shrinkage. These loads have been considered.

8.22.2 Required strength of the structure and its elements is shown in MATLAB.

8.22.3 Omit; Reference not Required

8.22.4 Modifications to the load combination equations that have been included include the following:

8.22.4.1 The load factor for the live load of this attraction can be found in MATLAB.

8.22.4.2 The impact factor stated in section 8.16 will be applied to the dynamic loads.

8.22.4.3 All redundant load combinations will not be applied in the calculations.

8.22.4.4 Joop Adventures is designed to withstand any amount of wind that can be encountered indoors. However, when the wind becomes too high of a speed, the ride will be closed until the speed lowers down to an acceptable speed for a certain amount of time.

8.23 *Material Allowables for Strength Using ASD:*

8.23.1 The materials used in the ASD analyses are appropriate and selected from an ASD reference.

8.24 - 8.26.1 Omitted since the Load and Resistance Factor Design (LRFD) is not being used.

8.27 *Design for Fatigue:*

8.27.1 The calculated working stress is used for the fatigue as shown in MATLAB.

8.27.2 The total number of the load cycles are applied in the fatigue calculations as shown in MATLAB.

8.27.3 Fatigue analysis has been calculated using a high cycle amount to calculate the maximum fatigue.

8.27.4 Fatigue evaluation has been consistent throughout the calculations.

8.27.5 Fatigue life was calculated and established during the design process of this ride.

8.27.6 There are some components that are excluded from the fatigue analysis, such as replacement parts and the box used for transportation.

8.28 *Load Factors for Fatigue:*

8.28.1 A fatigue load factor of at least 1.0 was used for the calculations as shown in MATLAB.

8.29 *Load Combinations for Fatigue:*

8.29.1 Multiple load combinations are used to produce the maximum stress fluctuation as shown in MATLAB.

8.30 *Fatigue Material Allowable Properties:*

8.30.1 A published fatigue property data, such as the S-N curve for copper tubing, is used for the fatigue design as shown in MATLAB.

8.30.1.1 The fatigue data that is based on mean data is reduced by at least two-standard deviations to account for material variability and reliability.

8.30.1.2 There is no fatigue property data that is derived from empirical data.

8.31 *Stability:*

8.31.1 Even though Joop Adventures is a portable ride, it will be stable even in every load during worst-case scenarios.

8.31.2 To inspect the ride for the stability of the ride, the inspector will visually see if the supports are still at the correct angle, see if the supports are properly connected, and the support bases are in the correct location. All the support should be standing in the support bases at a 90-degree angle except for one angled support at an angle of 45 degrees for the banked turn

8.32 *Metal Structures:*

8.32.1 Copper tubing is the only metal structure used for Joop Adventures. The copper tubing is used for the supports throughout the entire ride. The graph for the S-N curve of the copper tubing can be seen in MATLAB.

8.32.2 The mitigation for the copper tubing is stated in the following:

8.32.2.1 If there is any corrosion noticed on the outside of the support, the support will be replaced before any testing begins.

8.32.2.2 If there is any corrosion noticed on the inside of the support, the support will be replaced before any testing begins.

8.32.2.3 If there is any water noticed on the support, the support will be dried. If it causes corrosion, the support will be replaced.

8.33 *Timber Structures:*

8.33.1 The wooden structures for Joop Adventures include the base for the station, the base of the ride supports, and the box to enclose the PLC for theming. Pine wood was used for the PLC and the support bases. Whitewood was used for the station.

8.33.2 There will be no adjustments required for the loads and stresses of the wooden structures.

8.33.3 The wood will not put under enough stress to cause any damage.

8.33.4 There will be an inspection on every wooden structure. If there is any damage noticed, there will be a replacement part to replace it.

8.33.4.1 The inspection will mainly focus on looking for any rot, insects, or moisture damage.

8.33.4.2 This is omitted since there are no bolts and fasteners needed.

8.33.4.3 There will be an inspection for anything that may affect the load carrying capacity.

8.34 *Concrete Structures:*

8.34.1 This standard is omitted since there is no concrete used for manufacturing Joop Adventures.

8.35 *Plastic and Plastic Composite Structures:*

8.35.1 The loads and stresses of the 3-D printed PLA plastic track have been applied and shown in the Appendix, *Figures 11 – 16*, and for the calculations, please refer to the MATLAB code.

8.35.2 The track will have a snug fitted connecting between the track pieces, and between the bottom of the track and the copper support beams. These connections are snug enough to hold everything in place and not cause any splitting in the track when taken down for transport.

8.36 - 8.36.4 These standards are omitted since there is no fixed-air supported structure used for manufacturing Joop Adventures.

HYDRAULIC EQUIPMENT FOR AMUSEMENT RIDES AND DEVICES

9 – 9.16.3 – All standards for section 9 have been omitted as Joop Adventures does not contain any hydraulic equipment.

PNEUMATIC SYSTEMS AND COMPONENTS

10 – 10.3.6 - All standards for Section 10 have been omitted because Joop Adventures does not use or contain any devices that use pneumatic systems.

Safety Related Control Systems

11.2 *General Requirements*

11.2.1 The components that participate in hazard mitigation are as follows; four Adafruit Infrared Break Beam Sensors and one of the Elegoo Arduino Unos. This is because the break beam sensors are the only components that are used for monitoring ride car speed from one sensor to the other.

In the event of break beam sensor failure, the speed of the ride vehicle can no longer be monitored. This could lead to a potential crash. In the event of Arduino Uno, which monitors the sensor readings, the sensors would not be able to have their data read, which could lead to a potential crash if not resolved.

11.2.2 The break beam sensor is a component of hazard mitigation as can be seen in Figure 6.

11.2.2.1 In the event of any of the hazards identified in the ride analysis occurring, the break beam sensors will still be able to transmit data. This is assuming the following hazards have not occurred; system wide power failure, break beam sensor failure, Arduino, Basys 3, or Human-Machine Interface failure.

11.2.2.2 The break beam sensor is a component of hazard mitigation as can be seen in Figure 6.

11.2.3 The failure of the break beam sensors would significantly reduce the effectiveness of the SRCS. As a result, the following procedures occur.

11.2.3.1 Each time a break beam sensor detects the ride car passing it, it sends a time which is used to calculate the ride speed of the car. In the event of the break beam sensor failing, the time will not be sent meaning the speed will not be calculated, which would send an error to the Human-Machine Interface to notify the ride operator.

11.2.3.2 In the event of break beam sensor failure, the ride will undergo an automated forced stop procedure to get all riders safely off the ride so the attraction can undergo inspection.

11.2.3.3 When the break beam sensors are back up and running, the program will undergo one full loop of both ride cars going around the track before allowing riders back on to verify the failure no longer is present.

11.2.3.4 Mitigating the hazard is done automatically, refer to section 11.2.3.2.

11.2.4 Control functions are all centrally located. The Xilinx Basys 3 is responsible for assigning tasks to each Arduino Uno which then moves/senses its respective components.

11.2.5 Each operating mode will have a distinct operating procedure conducted by the Basys 3.

11.2.6 In the event of an external alarm, the rollercoaster will undergo the automated forced stop procedure as listed in 11.2.3.2.

11.2.7 With the exception of a potential crash, the SRCS is not responsible for the mitigation of other hazards such as deformation from the track supports as an example. This data can be found in the ride analysis.

11.2.8 When a different protocol occurs such as the automated forced stop, a warning will flash on the human-machine interface followed by a noise to grab the ride operator's attention.

11.3 Safety-Related Control Functions for Stopping

11.3.1 Joop Adventures does not comply to this section of the ASTM standards. This is because the ride analysis does not take into account the existence of a stop override. This is because to start the ride sequence of Joop Adventures, the start signal needs to be initiated.

11.3.2 Joop Adventures does not comply to this section of the ASTM standards. This is because the motors and sensors used for Joop Adventures are unable to be scanned for internal integrity. For example, if one of the servo motors failed internally, there would be no way to tell what is wrong with the servo motor aside from the fact that the component failed. This would result in a removal of the faulty component and replacement to a working component.

11.3.3 The operation referred to as an automated forced stop procedure in section 11.2.3.2 will be a preset option in the code to bring ride vehicles back into the station without proceeding up the lift hill. This allows the riders to disembark their respective ride vehicle safely as normal.

11.3.4 During the automated forced stop procedure, none of the equipment will impede the evacuation process of the patrons exiting the vehicle.

11.3.4.1 Due to the nature of the automated forced stop procedure, the patrons will be exiting the ride vehicles as normal. This results in a safe and orderly exit while remaining predictable and foreseeable to the ride operator.

11.3.4.2 Joop Adventures has no other alternative evacuation areas. This can be seen in the ride analysis.

11.3.4.3 Joop Adventures has no other unlisted evacuation areas. This can be seen in the ride analysis.

11.3.5 There are no individual mechanical mechanisms that dictate the initiation or ceasing of certain protocols. There are however, individual buttons created on the human-machine interface that can perform some of these tasks. The Ride Emergency Stop button causes a loss of all power to the system. This is a physical analog button, as well as being the only one used in Joop Adventures.

11.3.6 Manually Initiated Stopping Functions:

11.3.6.1 All manual controls reside on the human-machine interface except for the Ride Emergency Stop button. Each button can control the stopping of any function of the ride as well as the initiation of any part of the ride.

11.3.6.2 All of the ride protocols utilized for the functionality of Joop Adventures will all have the functionality of stopping the ride by using the automated forced stop procedure.

11.3.6.3 Every time the button responsible for starting the automated forced stop procedure is pressed, a temporary banner will appear on the screen to notify the ride operator that the button has been pressed.

11.3.6.4 Joop Adventures does not comply with this section because there is only one operator control terminal.

11.3.6.5 Every operator control for Joop Adventures will have the capability to stop the ride by means of automated forced stop procedure.

11.3.7 Joop Adventures meets the requirements of Section 11.3.7.1 and Section 11.3.7.2.

11.3.7.1 The automated forced stop procedure will bring the ride to a halt in the station, allowing the ride operator to leave the coaster unsupervised.

11.3.7.2 The automated forced stop procedure is designed to bring the ride to a stop as fast and safely as possible.

11.3.8 Ride Off Function:

11.3.8.1 The Ride Off Function is an emergency stop button that is hardwired before all components of Joop Adventures receive power. This component cannot fail.

11.3.8.2 Due to the intent behind the emergency stop button, serving as our "ride off function" in the event of a hazard or failure, the emergency stop button is intended to cut all power to Joop Adventures when toggled.

11.3.8.3 Joop Adventures does not comply with this section. This is because anybody who has access to the emergency stop button, which is our ride off function, is capable of releasing the button.

11.3.8.4 As indicated by figure 7, the ride off function is to be executed immediately when activated.

11.3.8.5 The ride off function resides next to the human machine interface, which is the main operator control.

11.3.8.6 In order to disable the ride off function, the ride operator would need to push down and twist the emergency stop button.

11.3.8.7 The ride off function is labeled as "Emergency Stop".

11.3.9 Ride Emergency Stop Function:

11.3.9.1 The ride emergency stop function is fail-safe as can be indicated in figure 7 of the ride analysis.

11.3.9.2

11.3.9.3 In order to disable the ride off function, the ride operator would need to push down and twist the emergency stop button.

11.3.9.4 The ride emergency stop function will always be adjacent to the human machine interface which sits in front of the ride operator.

11.3.9.5 Joop Adventures is designed to always have a ride operator in front of the human machine interface.

11.4 Safety-Related Control Functions for Starting:

11.4.1 Joop Adventures complies with both sections 11.4.1.1 and 11.4.1.2 for the following reasons;

11.4.1.1 Standard operating procedure will require initiation from the start button on the human machine interface that can be found on the screen.

11.4.1.2 The start button is designed to start Joop Adventures.

11.4.2 Joop Adventures does not comply with this section. This is because none of the protective measures such as safety harnesses can be checked via software. They can only be checked via visual confirmation by the ride operator.

11.4.3 The start function is initiated by a change of state from the start button. This start button signals a logical low to a logical high and then returns to a logical low after the signal is sent.

11.4.3.1 All acknowledgments of faults and alarms will not cause an accidental fault of the start function.

11.4.3.2 The start function is a function that will be distinct from the selection of operating modes.

11.4.4 The sequence of events needed to start any of the operating modes is enforced by the SRCS via operating mode selection.

11.4.5 Joop Adventures complies with the following sections as follows:

11.4.5.1 If a stop was caused by an operator, the code will not automatically begin any operation or protocol.

11.4.5.2 If a stop was caused by a fault or any automation, the code will not automatically begin any operation or protocol.

11.4.5.3 If a stop was due to a loss of power, the code will not automatically begin any operation or protocol.

11.4.5.4 If the previous stop placed the riders in a normal loading or unloading area, the code will not automatically begin any operation.

11.4.5.5 Even if the stop has resulted in the ride not operating for a longer than expected amount, the code will wait forever until told to begin an operation from the ride operator.

11.5 Operating Modes:

11.5.1 The human-machine interface has predefined operations that can be undergone automatically or be initiated by the ride operator. All of the changes in operating modes will appear on the screen.

11.5.2 Switching the operating modes of the ride will require a deliberate action by the ride operator to initiate on the human machine interface.

11.5.3 The ride operator is intended to remain in front of the human machine interface during all ride operations. This also restricts unauthorized persons from accessing the human machine interface.

Electrical Requirements

12.2 General Requirements for Electrical Installations

12.2.1 Wiring and Protection

12.2.1.1 This section does not apply as Joop Adventures cannot exceed 10 amperes.

12.2.2 Disconnects

12.2.2.1 Joop Adventures does not comply with this section since Joop Adventures does not have disconnecting locations aside from the Arduino Unos and Xilinx Basys 3.

12.2.2.2 This section does not apply as Joop Adventures does not have other voltage sources aside from the already listed DROK 48V power supply.

12.2.3 Grounding:

12.2.3.1 Joop Adventures is compliant with National Electrical Code, NFPA-70 as wires are hard enough, cables are intended to be exposed in sunlight, and our wires are far enough from the riders for this competition.

12.2.3.2 This section does not apply as Joop Adventures does not have switchboards or panel boards or any other type of enclosure.

12.3 Wiring Methods and Materials:

12.3.1 The only method of physical protection in the damaging of the wiring system is when the wires for sensors or power travel along the interior of the support columns and down to the foot of the base to travel above ground.

12.3.2 The electrical enclosure for the DC to DC Buck Converters do not meet the requirements for the appropriate rating.

12.3.3 This section does not apply as Joop Adventures does not have exposed electrical switches.

12.3.4 Joop Adventures does not follow the NEC standard and procedure for wiring systems.

12.4 Equipment for General Use:

12.4.1 This section does not apply as Joop Adventures does not have lighting fixtures.

12.4.1.1 This section does not apply as Joop Adventures does not have lighting fixtures.

12.4.1.2 This section does not apply as Joop Adventures does not have lighting fixtures.

12.4.1.3 This section does not apply as Joop Adventures does not have lighting fixtures.

12.4.1.4 This section does not apply as Joop Adventures does not have a portable cable.

12.4.2 This section does not apply as Joop Adventures does not have lighting fixtures.

12.4.3 Appropriate calculations were calculated for the duration of competition as well as the complete lifetime of the ride which includes operation for three years at two hours each week of those three years. For more specific information, please refer to the MATLAB code.

12.4.4 This section does not apply as Joop Adventures does not have transformers.

12.4.5 This section does not apply as Joop Adventures does not have power capacitors.

12.4.6 This section does not apply as Joop Adventures does not have collector ring or brush assemblies.

12.4.7 This section does not apply as Joop Adventures does not have grounding rings.

12.4.8 This section does not apply as Joop Adventures does not have equipment with temperature or humidity requirements.

12.5 Special Equipment:

12.5.1 This section does not apply as Joop Adventures does not have emergency lights.

12.5.2 Due to Joop Adventures being classified as SELV (Safety Extra-Low Voltage), the only wires that encounter metal, those which reside within the copper support tubes, do not need an extra ground wire.

12.5.3 This section does not apply as Joop Adventures does not have areas involving water or water vapor.

12.5.3.1 This section does not apply as Joop Adventures does not have wet areas.

12.5.3.2 This section does not apply as Joop Adventures does not have wet areas.

12.5.3.3 This section does not apply as Joop Adventures does not have wet areas.

12.5.4 Signs and Outline Lighting

12.5.4.1 This section does not apply as Joop Adventures does not have lighting fixtures.

12.5.4.2 This section does not apply as Joop Adventures does not have lighting fixtures.

12.5.4.3 This section does not apply as Joop Adventures does not have lighting fixtures.

12.6 Audio/Communication Systems:

12.6.1 Joop Adventures complies with this section because none of the speakers will be utilized to provide audio notifications for safety, operation, evacuation, or maintenance, of the ride.

12.6.2 Joop Adventures does not comply with this section because the cables used are not intended specifically for outdoor use or listed as UV resistant.

12.7 Portable Ride Assembly/Disassembly Conditions:

12.7.1 This section does not apply as Joop Adventures does not have cables that are at connecting points aside from where they are intended to connect to. This means that cables are not daisy chained.

12.7.2 During the assembly and disassembly of the ride, all power will be turned off and because there is no scenario where any component or element of the ride has the capacity to store charge, no energized points will be displayed to any personnel.

MECHANICAL SYSTEMS AND COMPONENTS

13.1 Scope:

13.1.1 This section pertains to all mechanical systems and components of Joop Adventures in relation to power transmission, patron-carrying devices, and general ride-safety devices.

13.2 Chain:

13.2.1 The chain used in the lift hill system for Joop Adventures was produced in compliance with the National Standard Institutes (ANSI) standard for chain-related devices

13.2.2 Through the ride analysis for Joop Adventures, the chain that was selected needed to be able to bear a load over 4 N of force, be able to handle speeds exceeding 100 mm/s, be made of proper material to combat corrosion and weathering and have the durability to be able to handle continuous operation. Refer to the MATLAB code for more specific calculations pertaining to the servo motor chain lift.

13.2.3 – The specification for this chain is as follows:

- (1) Dimension 9.7mm x 3.175mm
- (2) Load capacity – 4448 N
- (3) Pitch – 12.7mm
- (4) Tensile strength – 3188.90 g/mm
- (5) Strand – Single
- (6) Plate thickness – 12.7mm
- (7) Roller Diameter – 3.302 mm

13.2.4 The chain has been selected due to its compliance with ANSI standards.

13.2.5 The chain will pass through a total of three sprockets for the lift hill and has a factor of safety of 2427.1. For more information see the MATLAB code.

13.2.6 The chain exceeds the minimum requirements for the chain lift as stated in 13.2.5 as well as the MATLAB code.

13.2.7 Appropriate calculations were tabulated to ensure that the chain can support the ride vehicle with an acceptable factory of safety. See the MATLAB code for more detailed analysis.

13.2.8 There will be a guide on the lift hill to ensure that the chain maintains proper contact with the sprocket teeth and pulleys. Additionally, a sprocket will be used as a chain tensioner to ensure that the chain will not move in an unpredicted direction.

13.2.9 Appropriate maintenance measures for the chain was recorded in the maintenance section 5.7.1.3

13.2.10 Omitted as Joop Adventures does not feature a metallic chain guide. It features a plastic chain guide which will be appropriately lined to prevent excessive wear.

13.2.11 Cleaning and lubrication details can be found in section 5.7.1.3.

13.3 Wire Rope

13.3.1-13.3.17 Omitted as Joop Adventures does not have wire ropes or related accessories.

13.4 Anti-Rollback Devices

13.4.1 All of the sprockets on the chain lift hill feature anti-rollback devices that prevent the ride vehicle from falling back into the station after it latches onto the chain lift. Additionally, the dog chain on the underside of the ride vehicle prevents the ride from rolling back into the station.

13.4.2 Omitted as Joop Adventures includes anti-rollback devices.

13.4.3 See section 13.4.1 for explanation of current ride system as well as the MATLAB code for the calculations showing that Joop Adventures will comply with the appropriate load constraints.

13.4.4 Mechanical anti-rollback systems were described in section 13.4.1.

13.4.5 Omitted as the primary lift drive device features an anti-rollback precaution mechanism.

13.4.6 Joop Adventures features an anti-rollback system on the underside of the ride vehicle and on the primary lift drive.

13.4.7 Each individual Joop features an anti-rollback device, thus they will not be coupled together.

13.5 Machine Guards

13.5.1 Gates will be securely fastened around the station to ensure that riders do not enter prohibited areas within the station.

13.5.2 Omitted as parts were not tabulated to break free from the power transmissions.

13.5.3 Omitted as Joop Adventures does not feature drive shafts.

13.5.4 The lift hill chain for Joop Adventures is not enclosed with traditional guards. This design decision is justified by the ride's small scale, low vehicle mass (105 grams), and low operational speeds (0.1 m/s), which result in negligible risk of injury from chain contact.

Safety is maintained through the following measures:

Operator-Controlled Access: A single trained operator monitors all ride operations and ensures that passengers remain within designated ride zones.

Station Gates: Gates at the station prevent passengers from entering the lift hill or unauthorized areas where they could encounter the chain or sprockets.

Restricted Zones: The ride area around the chain and sprockets is clearly marked and inaccessible to riders during operation.

Maintenance Procedures: Operators perform all maintenance or inspection tasks with the lift hill stationary to eliminate any risk from moving chains.

These measures ensure that, despite the absence of a physical chain guard, the ride complies with the safety intent of ASME B15.1 by preventing operator or rider contact with moving chain components. Consequently, the design satisfies the operator safety requirements of ASTM F2291, section 5.4.1, through administrative controls, access restriction, and supervision.

13.6 Patron Lifting or Elevating Devices

13.6.1-13.6.5 Omitted as Joop Adventures does not feature hoists, power screw drives, and rack and pinion drives.

13.7 Brakes

13.7.1 General:

13.7.1.1 Joop Adventures features two friction wheels in the station to completely stop the ride vehicles and propel them out of the station. Additionally, Joop Adventures uses a magnetic braking system. In the case of an emergency, both ride vehicles will return to the station. Depending on the severity of the situation, one ride vehicle will stop along the lift hill, while the other ride will stop in the station until appropriate safety personnel can safely evacuate the riders.

13.7.1.2 The magnetic brakes used prior to the station serve to slow down the ride vehicle before it enters the station to prevent excessive deceleration as riders encounter the friction wheels. See MATLAB for calculations on the magnets to slow down the ride vehicle within the magnetic braking section.

13.7.1.3 The selection and design of the braking system is compliant with section 11 on Safety Related Control Systems.

13.7.2 Stopping and Safety Brakes:

13.7.2.1 The braking system, including neodymium magnets on the underside of the chassis and the track meet the specified needs of Joop Adventures as determined in the MATLAB calculations. 17 magnets will be used along the underside of the track to slow the ride vehicle down to an acceptable speed.

13.7.3 Retarding, Trim, or Reduction Brakes:

13.7.3.1 Brakes have been selected and tabulated according to Joop Adventure's specified standards. Refer to the MATLAB code for further information.

13.7.4 Parking Brakes:

13.7.4.1 The friction wheels within the station serve as parking brakes to prevent the ride vehicles from departing until the appropriate time interval.

14. Fencing, Guardrails, Gates, Walkways for Amusement Rides and Devices

14.1 All fencing, guardrails, gates, and walkways have been constructed according to the following requirements from standards 14.2 - 14.8. While gates are not a requirement of this competition, the team decided to implement this safeguard to manufacture a safe ride and challenge us to comply with industry standards.

14.2.1 Fencing, guardrails, and gates are designed and constructed to prevent guests from falling off the station.

14.2.2 Fencing, guardrails, and gates have been constructed according to the following:

14.2.2.1 The height of the fencing is 15 mm above the surface where the patrons stand.

14.2.2.2 Guardrails and fences have been constructed and placed around the station and stand a height of 15 mm above where the patron will be standing.

14.2.2.3 The spacing in the fence is constructed to meet the maximum number of spacing requirements.

14.2.2.4 The spacing in the fence on the stairs is constructed to ensure that riders cannot get through the gaps which are 5 mm in length and 10 mm in height.

14.2.2.5 This is omitted since there are no openings formed by radiused corners on the fences, guardrails, and gates.

14.3 Guardrails:

14.3.1 Guardrails in primary circulation areas comply with 14.2 and the following:

14.3.1.1 The guardrails have been constructed to resist a load from any direction applied to the top of the railings and side of the railings simulating riders climbing on top of the railing or leaning against it. The factor of safety is 7645.3. Further details on the calculations can be found in the MATLAB code.

14.3.2 - 14.3.2.3 This is omitted since there are no non-primary circulation areas that have been constructed.

14.4.1 - 14.4.1.6 This is omitted since there are no handrails due to this needing to be too small of a size.

14.5.1 The gates comply with 14.2 and the following:

14.5.1.1 Gates within the clearance envelope of the patrons comply with standard 6.6 according to the scope of this competition.

14.5.1.2 The entrance gates will open toward the ride, away from the patron waiting in line. The factor of safety for the gates supporting the weight of a rider pushing against the gate is 910.1. Further calculations can be seen in the MATLAB code.

14.5.1.3 The ride gates will open and close without contacting the patron or the ride.

14.6.1 The fences comply with standard 14.2 in the scope of this competition.

14.6.1.1 The fences will minimize the chance of injury between the spectator and rider, the spectator and ride components, and the rider and the fence.

14.6.1.2 The fences comply with the clearance envelope stated in standard 6.6.

14.6.1.3 The fences have been constructed to resist the force of the maximum weight of the ride vehicle and can be further explored in the MATLAB calculations.

14.7 This is omitted since there are no other devices for controlling guest flow.

14.8 There are walkways leading up to the rider gates and the exit gate.

14.8.1 This is omitted since there are no ramps that have been constructed.

14.8.2 This is omitted since there are no facility building codes to deviate from.

14.8.3 The elevated walkways have a fall protection system explained in 14.8.3.1.

14.8.3.1 The fall protection system used on Joop Adventures is stationary guardrails.

14.8.3.2 This is omitted since there are no temporary guardrails.

14.8.3.3 This is omitted since the fences and guardrails act as the structural barrier.

14.8.3.4 There is a fall protection for personnel on the elevated walkway leading to the rider gates. The fall protection system is stationary guardrails.

WELDING

15.1 - 15.4 All standards for section 15 have been omitted as Joop Adventures does not contain any welded parts in any of its main or sub-assemblies.

FASTENERS

16.1.1 The fastener that is used is a pin connection between each connected piece of 3-D printed PLA track. These fasteners have been considered in accordance with industry accepted engineering practices.

16.1.2 Track components are connected using custom 3D printed pin-style connectors manufactured from PLA material (100% infill). While these connectors are not commercially standardized ANSI/SAE hardware, they were designed and verified in accordance with accepted engineering practices.

The maximum calculated shear load per connection under governing dynamic load conditions is less than 0.3 N. This load is significantly below the allowable shear strength of the PLA material, resulting in an adequate factor of safety.

Pins are installed with full engagement and positive geometric retention within the track joint geometry, preventing unintended disengagement during operation.

Therefore, although custom-fabricated, these fasteners meet the intent of ASTM F2291 requirements for fasteners conforming to accepted engineering standards through strength verification, material validation, and inspection procedures.

16.1.3 The 3D printed pin-style connectors are custom-fabricated components and are not standardized commercial fasteners; therefore, grade marking in accordance with ANSI/SAE or ASTM bolt standards is not applicable.

Material identification is controlled through documented filament specification (PLA) and internal design documentation. Strength adequacy is verified through engineering analysis and inspection procedures.

16.1.4 The exact specification of all fastening components is defined within the engineering documentation for the ride system.

For 3D printed PLA pin connectors, documentation includes:

Material: PLA (Polylactic Acid)

Infill: 100%

Print orientation to avoid shear along layer lines

Nominal pin diameter and engagement length

Calculated shear loading under maximum dynamic design loads

All fastening components have been verified through engineering analysis to ensure stresses remain below allowable material limits with appropriate factor of safety.

16.1.5 Through-bolting is the preferred connection method for structural assemblies where metallic hardware is utilized.

In this design, primary track joints are assembled using integrated 3D printed pin-style connectors due to the small-scale and modular nature of the ride system. These connections provide full engagement through the joint interface and are not intended for frequent removal during normal operation.

The use of custom printed pins is deemed appropriate by the designer/engineer based on calculated loading conditions, which are significantly below material capacity.

16.1.6 The design avoids placing threaded fasteners in primary shear wherever possible.

The 3D printed pin connectors function as smooth, non-threaded shear pins. Where shear loading exists, the connection geometry provides double-shear conditions through symmetric joint engagement.

Calculated maximum shear load per connection under governing dynamic conditions is less than 0.3 N, which is substantially below the allowable shear strength of PLA material.

No threaded metallic fasteners are subjected to primary shear loading.

16.1.7 The 3D printed PLA pin-style connectors are non-tensioned components and do not require torque application. They are installed by full geometric engagement within the track joint.

Since no tensioned bolts, nuts, or washers are used in primary load-bearing applications, torque specifications are not applicable.

Where auxiliary metallic fasteners are present (if applicable), installation follows manufacturer torque recommendations consistent with accepted engineering practice.

16.1.8 A. Fastener Type and Location

Primary structural fasteners consist of custom 3D printed PLA pin-style connectors used at track-to-track joints.

These pins provide geometric retention and operate in shear under dynamic loading conditions.

No threaded metallic fasteners are used in primary load-bearing track joints.

B. Material Identification

Material: Polylactic Acid (PLA)

Infill: 100%

Fabrication Method: Fused filament fabrication (FFF)

Nominal diameter and engagement length are specified in design drawings.

Material specification documentation is maintained as part of the engineering design file.

C. Inspection Requirements

The following inspections shall be performed:

1. Pre-Operation Inspection (Before Each Operating Period)

Verify full engagement of each pin.

Check for visible cracking, whitening, or stress marks.

Confirm no axial displacement or loosening.

Ensure no deformation at shear interfaces.

2. Periodic Inspection (Every 50–100 Operational Cycles)

Remove and visually inspect representative sample pins.

Inspect joint bores for wear or elongation.

Perform light manual retention check.

3. Annual Inspection

Inspect all pins for wear, creep, or fatigue damage.

Replace any pin showing dimensional reduction, cracking, or deformation.

Verify shear surfaces remain smooth and undamaged.

D. Acceptance and Rejection Criteria

A fastener shall be removed from service if any of the following conditions are observed:

Visible crack formation

Layer delamination

Permanent bending or distortion

Surface whitening indicating plastic deformation

Looseness preventing full engagement

Measured dimensional reduction greater than 5%

Replacement pins shall meet original material and dimensional specifications.

E. Replacement Procedure

Replacement pins shall be manufactured using identical PLA material and 100% infill.

Print orientation shall match original design intent to avoid shear along layer lines.

Replacement components shall be visually inspected prior to installation.

F. Documentation

Inspection and replacement records shall include:

Date of inspection

Number of operational cycles (if applicable)

Observed condition

Replacement actions taken

Inspector name

Records shall be retained as part of the ride maintenance documentation.

16.1.9 SAE Grade 8 fasteners that have been previously torqued to a value greater than 75% of the ultimate strength will not be reused.

16.2.1 This is omitted since there are no washers being used.

16.2.2 This is omitted since there are no washers being used.

16.2.3 This is omitted since there are no washers being used.

16.2.4 This is omitted since there are no washers being used.

16.3

16.4.1 This is omitted since there are no locking systems used as fasteners.

16.4.2 This is omitted since there are no locking systems used as fasteners.

16.5.1 This is omitted since there are no nuts and bolts used as fasteners.

16.5.2 This is omitted since there are no materials being clamped.

16.5.3 This is omitted since there are no materials being clamped.

OPERATOR CONTROLS

17.1 Because the ride operator is intended to remain in front of the ride operator control module, that being the human machine interface for all operations of the ride.

17.2 The operator controls system is created with the intention of every action being a deliberate and distinct action which results in a lack of unintentional activations.

17.3 Operator Controls are all identified in the English language.

17.4 Operator Controls will require a pin before the initial start-up sequence can begin for the system. This ensures that random people cannot gain access to the system.

DOCUMENTATION

18.1 There is supporting documentation that can be found in the following subsections that will cover sections from Practice F1193:

18.1.1 *Section 11 Operational Instruction Requirements:*

11.1 In compliance with ASTM F1193 standards, all operation instructions will be presented within the English language for the purpose of harmonization.

11.1.1 For all documentation of Joop adventures ride operation, including all major and minor components, please refer to the theory of operations

11.1.1.1 The ride will begin in the station. After being dispatched from the station, patrons will be lifted 417 mm above the station and 445 mm above the baseplate on a 90-degree lift hill. Following this lift, they will experience a small drop before turning to into an inclined straight, followed by a turn into a downward helix. As they leave the helix, they will enter a 45-degree outer bank, then turn into a magnetic braking section before re-entering the station.

11.1.1.2 load/unload procedures will be as follows. As the ride vehicle enters the station and comes to a complete stop, the ride vehicle restraints will be raised as the exit gates open, and existing patrons will be asked to quickly and carefully exit the ride. During this time, the loading gates will open, and boarding guests will be told to move to the most available seat. Once seated, ride operators will pull down the lap bar and the over-the-shoulder restraint. Once all restraints have been locked operator will begin the dispatch verification and then start the ride.

11.1.2 The following is a comprehensive detail of the safety requirements by ride operators to ensure their own and patrons' safety.

11.1.2.1 Maximum capacity is 4 riders per ride vehicle, with each rider's weight no greater than 5 grams. Exceeding this weight may compromise the 15% infill structural integrity.

11.1.2.2 The types of restraints are a ratcheted lap bar and an over-the-shoulder restraint. Each restraint will be pulled down by a member of ride operations, with the lap bar positioned against the patron's lower body and the over-the-shoulder restraint clamped into place to protect the patron's upper torso.

11.1.2.3 Before each cycle, the operator must perform a visual inspection of all restraints

11.1.2.4 During boarding of Joop Adventures, riders will be told to remove all loose articles before entering the ride, and told to be seated and raise their arms until a member of ride operation comes to secure their harness

11.1.2.5 Do not operate above 104°F to prevent thermal softening of the 3D-printed filament.

11.1.3 This section shall cover ride inspection by maintenance members & ride operation members.

11.1.3.1 Prior to park opening, the following inspection must be made by ride operators or members of maintenance:

1. Check for the elongation of the pin holes in the track.
2. Visual and physical inspection of the track to ensure no delamination has occurred.
3. Inspect the operator controls to make sure there are no malfunctions.
4. Visually inspect all footers to make sure there is no splitting or uneven deformation.
5. Visual Inspection of the lift hill chain to make sure there are no deformities or broken links in the chain.
6. Inspection of the ride platform to ensure there are no potential dangers that could cause injuries to ride operators or patrons
7. A visual four cycle inspections per will be conducted to ensure all ride components are operational
8. A final on-ride inspection will be conducted by a member of ride operations to ensure there are no obstructions that could cause any problems with

11.1.3.2 At least one ride operator will be stationed behind the operator panel at all times to monitor ride operations, and during loading/unloading procedures, Other operators will be on the platform perform restraint checks

11.1.3.3 For this compliance, refer to the theory of operations.

11.1.4 This section shall give a brief overview of all emergency plans for Joop adventures

11.1.4.1 For emergency procedures, all operational components will go into the automated emergency stop. During this time, all power will be cut to Joop Adventures, and all trains will be sent to the nearest safety zone so that they may be safely placed in a location that ride operator(s) & emergency services can easily access to remove riders from Joop Adventures.

11.1.4.2 The use of an emergency stop button will be implemented for swift control of the ride in case of emergencies.

11.1.4.3 The emergency stop button is a mushroom-shaped device with a labeled outer ring and a spring-loaded red top.

11.1.4.4 For a full description, refer to section 12 of this document

18.1.2 Section 13 Maintenance Procedure Requirements:

13.1 This section shall give an overview of all maintenance instructions required by the owner of the ride for the purpose of harmonization. All instructions shall be in the English language.

13.1.1 For a full description of all Joop adventures and its major and minor components, please refer to sections 7–12 of this document and to the theory of operations for additional information.

13.1.1.1 For the designed motion description of Joop Adventures, please refer to the theory of Operations

13.1.2 All components of Joop Adventures will be properly labeled with a numbered system that will help with the installation of all ride components, starting from the baseplate to the final piece of track.

13.1.3 In accordance with ASTM standards, a proper plan must be made to ensure the lubrication of specific ride components after an extended period of time.

13.1.3.1 Apply dry or wet to vehicle axles and wheels to maintain consistent friction coefficients. Also, to the sprocket and lift hill chain to avoid corrosion and weathering

13.1.3.2 Lubricant should be applied to these sections every other day.

13.1.3.3

Ride Vehicle Application Procedure:

1. Remove the vehicle from the track.
2. Apply a single drop or puff of lubricant directly to the interface where the axle meets the wheel hub.
3. Spin the wheel manually for 30 seconds to ensure even distribution.
4. Wipe away all excess with a lint-free cloth. No visible wetness should remain.

Carbon Steel Lift Hill Chain Application Procedure:

1. Place a paper towel underneath the lift hill to protect the 3D-printed track.
2. Run the lift hill motor at its slowest speed.
3. Apply the lubricant to the inside of the chain loops as it passes over the top sprocket.
4. Allow the chain to complete three full revolutions.
5. Use a cloth to "cycle" the chain through your fingers to remove excess oil, leaving only a thin, protective film.

I.3. Sprockets (Lift Hill) Application Procedure:

1. Axle/Bore: Apply a small dab of Lithium grease to the shaft before sliding the sprocket on.
2. Teeth: Do not use heavy grease on the sprocket teeth; it will fly off (sling) at high speeds. Use a dry-film lubricant to reduce chain-tooth engagement noise.

3. Inspection: Check that no lubricant has migrated from the sprocket to the 3D-printed support structure.

13.1.3.4 Refer to the previous section.

13.1.4 Refer to section 11.1.3.1 of section 18 for all pre-daily inspections.

13.1.5 This section shall provide a brief overview of the frequency of daily pre-park open inspections that are needed for tolerances.

13.1.5.1 In accordance with ASTM F1193, recommended wear limits and tolerances have been established for all safety-critical and motion-related components. Components subject to wear include rolling bearings (17×23×4 mm), track running surfaces, 3D-printed PLA support interfaces, servo mounting brackets, and mechanical fasteners within the gondola and station assemblies.

Rolling element bearings shall be replaced if radial play exceeds 0.25 mm, if surface pitting is visible, or if rotational resistance increases beyond normal operating torque. Track running surfaces shall not exhibit wear greater than 1.0 mm material loss or deformation exceeding 2% of nominal cross-sectional thickness. Fasteners shall be replaced if thread wear, corrosion, or elongation is observed, or if torque retention falls below 90% of specified installation torque.

3D-printed structural PLA components shall be removed from service if crack initiation, layer delamination, or dimensional distortion greater than 1.5 mm from nominal geometry is measured.

All wear measurements shall be verified using calibrated calipers or micrometers during scheduled inspections. Components exceeding the recommended wear limits shall be immediately removed from service and replaced prior to continued operation.

13.1.5.2 To ensure continued operation within prescribed operational limits, the following operational tests shall be performed by the owner/operator at the minimum intervals specified below.

1. Pre-Opening Daily Inspection (Before Each Operating Day)

Purpose: Verify system readiness and safety-critical functionality.

Visual inspection of track, support columns, gondola, and station components for visible damage, deformation, cracking, or loosened fasteners.

Manual verification of lap restraint engagement and locking function.

Functional verification of servo motors (MG90S positional and FS90R continuous) to confirm proper response and full range of motion.

Verification of infrared break beam sensor alignment and detection response.

Emergency stop push button (LAY7) activation test to confirm immediate power interruption.

Operational Limit:

All systems must respond within programmed parameters and show no structural deformation or abnormal noise.

Action if Failed:

Ride shall not operate until deficiency is corrected.

2. Weekly Functional Load Test

Purpose: Verify structural and mechanical integrity under representative loading conditions.

Apply simulated operational load equivalent to maximum design load (including gondola and passenger mass equivalent).

Observe track deflection and support column response.

Confirm lap restraint maintains secure locking under applied load.

Operational Limit:

Measured deflection shall not exceed calculated design deflection values by more than 10%.

No permanent deformation shall occur.

Action if Failed:

Component must be removed from service and inspected for structural compromise.

3. Monthly Mechanical and Electrical Verification

Purpose: Confirm continued compliance with design tolerances.

Measure bearing radial play (replace if >0.25 mm).

Verify fastener torque (must retain $\geq 90\%$ of specified torque).

Confirm servo positional accuracy within $\pm 2^\circ$ of programmed setpoint.

Verify power supply voltage output remains within $\pm 5\%$ of rated value.

Operational Limit:

All measurements must remain within prescribed tolerances established in the design documentation.

Action if Failed:

Affected subsystem shall be recalibrated, repaired, or replaced prior to operation.

4. Annual Comprehensive Inspection (or End-of-Competition Cycle)

Purpose: Full system integrity verification.

Remove and inspect critical load-bearing components.

Inspect 3D printed PLA components for internal cracking or delamination.

Re-validate safety factor assumptions if cumulative cycle count exceeds design estimate.

Operational Limit:

No fatigue cracking, material degradation, or dimensional distortion beyond established limits may be present.

Action if Failed:

Component replacement required before continued service.

Documentation Requirement

All inspections and tests shall be documented in a maintenance log including:

Date of inspection

Inspector name

Measured values

Corrective actions taken

Records shall be retained for the duration of ride operation.

13.1.5.3 1. Visual Inspection (VT)

Components:

Track running surfaces

PLA support columns and 3D-printed structural members

Lap restraint arms and pivot joints

Mechanical fasteners

Gondola frame and mounting brackets

Areas of Focus:

Stress concentration regions

Layer interfaces (for 3D-printed components)

Bolt holes and fastener interfaces

Frequency:

Daily pre-operation inspection

Detailed visual inspection monthly

Acceptance Criteria:

No visible cracking, layer delamination, or material separation

No permanent deformation exceeding 2% of nominal geometry

No loose or elongated fastener holes

No corrosion or wear beyond prescribed limits

Rejection Criteria:

Any crack propagation, structural distortion, or fastener compromise requires removal from service.

2. Dimensional Verification (Caliper/Micrometer Measurement)

Components:

17×23×4 mm rolling element bearings

Track thickness at high-contact zones

Support column outer diameter

Lap restraint pivot pin diameter

Areas of Focus:

Bearing race play

High-friction or rolling contact areas

Pin and joint interfaces

Frequency:

Monthly

After any abnormal operational event

Acceptance Criteria:

Bearing radial play ≤ 0.25 mm

Track wear ≤ 1.0 mm material loss

Pivot pin wear $\leq 5\%$ diameter reduction

Dimensional distortion ≤ 1.5 mm from nominal geometry

3. Dye Penetrant Testing (If Metallic Components Are Present)

Components:

Metal support inserts (if used)

Load-bearing bracket

Lap restraint metal arms or pins

Areas of Focus:

High bending stress regions

Pin transitions and shoulder fillets

Frequency:

Annually

After overload or forced stop incident

Acceptance Criteria:

No linear indications exceeding 3 mm in length

No clustered porosity or crack-like indications

4. Functional Electrical Verification (Non-Destructive System Testing)

Components:

Infrared break beam sensor

Servo motors (MG90S, FS90R)

Emergency stop push button

Power supply output

Areas of Focus:

Signal response time

Positional accuracy

Voltage stability

Frequency:

Daily functional test

Monthly calibration verification

Acceptance Criteria:

Servo positional accuracy within $\pm 2^\circ$ of programmed value

Power supply output within $\pm 5\%$ rated voltage

Emergency stop must cut power within immediate activation (no perceptible delay)

Sensor detection response within programmed threshold

Documentation and Corrective Action

All NDT results shall be recorded in the ride maintenance log. Any component failing to meet the acceptance criteria shall be:

Removed from service immediately

Evaluated for repair or replacement

Re-tested prior to return to operation

13.1.5.4 Tests will be compliant with the following criteria:

(1) Tests will be performed prior to competition.

(2) Tests will be passed during the ride lifetime.

(3) Tests will be created in accordance that the operator can effectively complete them.

13.1.6 Only 3D-printed pins matching the original file specifications (100% infill recommended for pins) or specified plastic hardware may be used.

13.1.7 This section shall cover all electrical power schematics.

13.1.7.1 Refer to Appendix *Figure 35* for the electrical power schematic.

13.1.7.2 Omit. Joop Adventures does not comply with this standard.

13.1.7.3 Omit- Joop Adventures does not have any with numbered identification

13.1.8 This section is omitted because Joop Adventures does not contain any Pneumatic components or systems

13.1.8.1 Omitted

13.1.9 Refer to the drawing package for all part assemblies.

13.1.10 In the event of no Operation Joop Adventures will be placed under ride delay until further analysis has been conducted to open to the public.

13.1.11 All components shall be disassembled starting from the uppermost track and all following numbered components

13.1.12 The only restriction to lubrication is the use of WD-40, as the chemicals in that lubricant can degrade the materials in the 3D-print filament

13.1.13 At this time Joop Adventures does not adhere to this standard as no special manufacture serial number will be made for this ride

18.1.3 Section 15 Inspection Requirements:

15.1 1. General

Inspection procedures for the ride shall be performed in accordance with ASTM F1193 and ASTM F770, ensuring safe operation of mechanical, structural, electrical, and control systems. Inspections shall be conducted prior to each operation period and periodically as required for routine maintenance or after any abnormal events.

The purpose of these inspections is to identify:

Structural damage or wear

Fastener loosening or withdrawal

Mechanical misalignment

Electrical or control system malfunctions

Any condition that could compromise patron safety

2. Pre-Operation Inspection

Before ride operation, the following items shall be inspected:

Structural Components:

Check support columns, baseplates, and track for cracks, deformation, or wear.

Verify that all 3D-printed and plywood components are intact and properly seated.

Inspect any fasteners or connector pins for full engagement.

Mechanical Systems:

Inspect the lift mechanism, chains, sprockets, and drive components for alignment and free movement.

Verify that all moving parts operate smoothly and without obstruction.

Control and Electrical Systems:

Verify HMI interface functionality.

Test all sensors (e.g., infrared break beam) for correct operation.

Confirm that emergency stop devices function properly and interrupt power.

Safety Devices:

Check automated forced stop procedures for proper response.

Ensure any restraint or containment devices are intact and functional.

3. Operational Inspections During Use

Observe ride operation for abnormal sounds, vibrations, or binding.

Verify that vehicle motion remains within design velocity and acceleration limits.

Confirm no lateral or vertical displacement exceeds design tolerances at support columns.

4. Periodic Detailed Inspections

Conduct a thorough visual inspection weekly or after a set number of operating cycles (e.g., 500 cycles).

Check all fasteners, pins, and connectors for wear or fatigue.

Inspect painted/finished surfaces for degradation, moisture damage, or biological growth.

Verify that all electrical wiring and connections remain secure and free of damage.

5. Documentation and Reporting

Each inspection shall be documented, including date, inspector name, and any deficiencies identified.

Any damaged or compromised components must be repaired or replaced before resuming operation.

Inspection records shall be retained in accordance with ASTM F1193 requirements.

6. Compliance Statement

This inspection procedure follows the practices outlined in ASTM F1193 and ASTM F770, adapted for the scale and construction of the current ride system. All inspections are designed to ensure continued structural integrity, mechanical function, and electrical safety.

15.1.1 Any information within 15.1 that is deemed essential by the manufacturer at the time of delivery shall be communicated to all known owner/operators.

15.2 All inspections will meet the following requirements:

15.2.1 Inspections will be performed by the manufacturer prior to attending the Ride Engineering Competition.

15.2.2 Inspections include elements of the ride that can be reasonably expected to pass throughout the lifetime of the ride.

15.2.3 Inspections are reasonable and appropriate for the operator to perform.

15.3 If there is an incident involving a critical component the manufacturer will promptly evaluate the information and disseminate their findings to the known owner/operator.

18.1.4 Section 12 Testing Performance Requirements:

12.1 All of the following tests will be performed on the ride prior to the Ride Engineering Competition. Constructing a prototype on this scale was outside of the bounds of the current timeframe, so the team decided to conduct these tests on the current ride and change or replace any components that failed to meet the adequate requirements.

12.1.1 Procedures to Verify Maximum Safe Design Loads:

12.1.1.1 Manual inspection will be conducted prior to the ride starting to ensure that the ride has not significantly deflected to affect the ride experience and that the loads placed on the other components of the ride assembly have not caused sufficient damage to affect the ride experience. Additionally, calculations have been tabulated in the provided MATLAB and excel files to demonstrate the lifetime of the ride as well as the forces experienced.

12.1.1.2

12.1.1.3 Visual measurements of the ride to ensure that the components are within the specified tolerances and fit the current rider specifications included in the Ride Engineering Prompt [1] and Figure 3.

12.1.1.4 Any other procedures to ensure that the ride is operating normally can be viewed in the risk assessment, ride analysis, and hazard mitigation sections in (*Appendix 4-10*) below.

12.2 Installation Testing Requirements

12.2.1 Our team will provide the following standard testing guides:

12.2.1.1 Materials Testing:

1. General: Material selection and verification were performed in accordance with the intent of ASTM F1193 to ensure all structural and non-structural components are suitable for their intended use, properly rated for applied loads, and free from defects that could compromise safety.

Testing and verification methods included manufacturer data review, visual inspection, dimensional verification, load analysis comparison, and functional testing where applicable.

2. Copper Support Tubes

Material: Type M Copper Tubing

Verification Methods:

Manufacturer specification review for yield strength and wall thickness.

Dimensional measurement of outer and inner diameters.

Visual inspection for dents, kinks, or corrosion.

Structural analysis comparison between calculated maximum loads and allowable stress.

Acceptance Criteria:

Calculated stresses under maximum dynamic loading remain below allowable material stress with a factor of safety > 2 .

No visible deformation or instability under applied load.

3. Wooden Base Plate and Station Structure

Material: Yellow Pine (structural grade plywood and lumber)

Verification Methods:

Visual inspection for cracks, delamination, knots in critical areas, or moisture damage.

Dimensional verification of thickness (5/16 in plywood).

Load distribution analysis to verify compressive stress under footings.

Stability verification against sliding and overturning.

Acceptance Criteria:

No structural defects in load-bearing regions.

Calculated compressive stresses below allowable wood stress values.

Structure remains stable under maximum lateral load case.

4. 3D Printed Track

Material: PLA

Verification Methods:

Dimensional verification of cross-section.

Visual inspection for layer separation or print defects.

Static load testing using equivalent vehicle weight and dynamic load multiplier.

Comparison of simulated stresses (SolidWorks/ANSYS) to PLA tensile strength.

Acceptance Criteria:

No cracking, delamination, or permanent deformation.

Calculated stress remains below allowable PLA strength with acceptable factor of safety.

5. 3D Printed Ride Vehicle

Material: PLA

Verification Methods:

Mass verification (0.105 kg).

Visual inspection for cracks or print defects.

Wheel/axle alignment inspection.

Functional rolling test through full track circuit.

Acceptance Criteria:

No structural degradation after repeated cycle testing.

Proper engagement with track and lift system.

6. Foam Volcano (Thematic Element)

Material: Lightweight Foam

Verification Methods:

Visual inspection for detachment risk.

Secure mounting verification.

Clearance verification from track envelope.

Acceptance Criteria:

Non-load-bearing component securely attached.

No interference with ride envelope.

7. Fog Machine

Verification Methods:

Manufacturer electrical rating verification.

Functional testing for proper operation.

Thermal clearance inspection.

Electrical connection inspection.

Acceptance Criteria:

Operates within rated voltage and current.

No overheating observed.

No moisture accumulation affecting structural components.

8. Speaker System

Verification Methods:

Electrical rating review.

Wiring inspection.

Functional audio test.

Acceptance Criteria:

No exposed wiring.

Stable mounting.

Operates within rated electrical parameters.

9. Laboratory and Station Systems

Verification Methods:

Structural inspection of station walls and supports.

Electrical wiring inspection per low-voltage safety practices.

Emergency stop functionality testing.

Acceptance Criteria:

All structural members secure.

Electrical systems properly insulated.

Emergency stop fully interrupts power when activated.

Summary Statement

All materials and components were verified through inspection, manufacturer data review, load analysis comparison, and functional testing in accordance with the intent of ASTM F1193. Structural components demonstrate adequate strength

under maximum dynamic design loads, and non-structural elements are securely mounted and do not interfere with ride operation.

12.2.1.2 Erection/Modification Acceptance Testing: Upon completion of assembly, the following inspections and tests shall be performed:

2.1 Structural Verification

Confirm all 40 support columns are properly seated on the wooden base plate.

Verify copper support tubes are plumb and properly connected.

Inspect 3D printed track sections for proper alignment and secure attachment.

Confirm plywood base plate (5/16 in thickness) is flat and stable.

Verify station and thematic structures (lab, volcano) are securely mounted.

Acceptance Criteria:

No visible misalignment, looseness, cracking, or instability under static loading.

2.2 Mechanical System Testing

Verify lift hill alignment (including 90° lift section).

Confirm smooth engagement of vehicle with lift mechanism.

Perform manual rotation check of moving components.

Verify free rolling of vehicle along entire track circuit

Acceptance Criteria:

No binding, derailment, or abnormal vibration observed.

2.3 Electrical and Control System Testing

Verify proper wiring connections.

Power-on test of control system and HMI.

Test infrared break beam sensors for proper detection.

Confirm operation of fog machine and speaker system.

Verify emergency stop button interrupts system power immediately.

Acceptance Criteria:

All control and safety devices function as intended without delay or malfunction.

2.4 Functional Load Test

Operate vehicle through full track circuit under normal conditions.

Conduct multiple consecutive cycles (minimum 5 full cycles).

Observe for excessive deflection, vibration, or abnormal noise.

Verify structural stability under maximum dynamic loading conditions.

Acceptance Criteria:

Ride operates continuously without structural deformation, fastener loosening, or system fault.

3. Modification Acceptance Testing

If any structural, mechanical, electrical, or control modification is made, the following steps shall be performed prior to returning the ride to service:

Inspect modified component for proper installation.

Verify compatibility with adjacent systems.

Re-perform applicable structural and functional tests.

Conduct minimum three successful full operational cycles.

Re-verify emergency stop functionality.

Acceptance Criteria:

Modified system performs equal to or better than original acceptance condition and maintains compliance with ASTM F1193 safety intent.

4. Documentation

All erection and modification tests shall be documented.

Record date, description of work performed, inspector name, and test results.

Any deficiencies shall be corrected prior to operation.

Compliance Statement: Erection and modification acceptance testing has been developed in accordance with the intent of ASTM F1193 to ensure structural integrity, mechanical functionality, and operational safety prior to service.

12.2.1.3 Performance testing: Prior to performance testing:

Erection acceptance testing shall be completed.

All inspections per ASTM F1193 and F770 shall be verified.

Emergency stop systems shall be confirmed operational.

All structural components shall be visually inspected.

3. Operational Cycle Testing

3.1 Full-Circuit Operational Test

Operate the ride vehicle through the complete track layout, including the 90° lift hill.

Perform a minimum of 10 consecutive full cycles.

Observe vehicle motion, speed consistency, and track engagement.

Acceptance Criteria:

No derailment or loss of track engagement.

Smooth transition through curves and lift sections.

No excessive vibration or abnormal noise.

3.2 Maximum Dynamic Load Verification

Monitor structural response during peak loading locations.

Confirm no permanent deformation of copper support columns.

Verify no cracking or delamination in 3D printed track sections.

Observe baseplate stability under maximum lateral load condition.

Acceptance Criteria:

No visible structural deformation.

Support columns remain plumb.

Baseplate does not shift or slide.

3.3 Control System Performance Test

Verify proper sensor triggering (infrared break beam).

Confirm automated stop procedure activates when required.

Test emergency stop activation during operation.

Acceptance Criteria:

Immediate interruption of drive system upon E-stop activation.

Correct detection and response from all sensors.

System resets properly after stop condition.

3.4 Thematic and Auxiliary System Verification

Operate fog machine during ride cycle.

Verify speaker system functions continuously without electrical fault.

Confirm decorative elements (volcano, lab structures) remain secure.

Acceptance Criteria:

No interference with ride envelope.

No electrical overheating or loose wiring.

4. Stability and Safety Observation

During testing, observe:

Lateral stability of support columns.

Any baseplate movement.

Fastener loosening.

Excessive oscillation or vibration amplification.

If any unsafe condition is observed, testing shall be stopped immediately and corrective action taken before resuming.

5. Documentation

The following shall be recorded:

Date of testing

Number of cycles completed

Observations

Any corrective actions taken

Name of responsible tester

Records shall be retained as part of the ride's safety documentation.

Compliance Statement: Performance testing has been conducted in accordance with the intent of ASTM F1193 to verify that the ride operates within its designed structural, mechanical, and safety parameters under expected operating conditions.

12.3: Operational Resting Requirements:

12.3.1: Our team shall develop operational tests stating minimum intervals for the tests to determine whether a given ride or device is operating within prescribed operational limits. See *Appendix Figures 4-10* below for Risk Assessment, Ride Analysis, and Hazard Mitigation.

12.3.2 All tests will be provided prior to the competition with sufficient evidence that Joop Adventures complied with an acceptable margin.

12.3.2.1 All tests will have satisfactory performance prior to the competition.

12.3.2.2 Joop Adventures will be able to complete these tests within the lifetime of the ride, assuming recommended maintenance and operational procedures are followed.

12.3.2.3 All tests will be reasonably completed by the owner/operator.

12.3.2.4 Any operational testing including load testing performed on an amusement ride or device shall be completely nondestructive in nature. Overloading is deemed inappropriate.

12.3.2.5 All installation and testing will be conducted within the stated limits of the information displayed in the MATLAB and Excel calculations which depend on limits of the materials utilized.

12.4 Non-Destructive Testing Requirements:

12.4.1 These tests will be performed in accordance with the components limits displayed in the MATLAB and Excel data. Some of these tests include, but are not limited to, the following:

1. Visual Inspection (VT)

Applicable To:

Copper support tubes

3D printed track

3D printed vehicle

Wooden baseplate and station

Fasteners and connections

What to Check:

Cracks, deformation, or bending

Layer separation in PLA prints

Splitting or delamination in plywood

Fastener loosening

Corrosion or denting in copper tubes

This is the primary and most important NDT method for small-scale systems.

2. Dimensional Verification

Applicable To:

Track alignment

Column verticality

Baseplate flatness

Lift hill geometry

Method:

Calipers

Ruler

Square

Level

Ensures no structural shifting or warping has occurred.

3. Deflection Testing Under Load (Static Load Test – Non-Destructive)

Applicable To:

Track sections

Support columns

Lift hill

Method:

Apply known load equal to vehicle weight $\times$ impact factor.

Measure vertical or lateral deflection.

Compare to initial baseline measurement.

No permanent deformation should occur.

4. Tap Test (Acoustic Sounding)

Applicable To:

3D printed PLA parts

Wooden baseplate

Station structures

Method:

Lightly tap surface with small tool.

Listen for hollow or dull changes in sound.

Changes may indicate internal cracking, delamination, or voids.

5. Fastener Torque/Retention Check

Applicable To:

Bolted copper connections

Mechanical mounts

Structural joints

Method:

Verify hand-tight or torque specification.

Ensure no rotational movement.

Confirms no loosening from vibration.

6. Operational Vibration Observation

Applicable To:

Entire structure during full cycle

Method:

Run multiple cycles.

Observe oscillation amplitude at support columns.

Check for resonance behavior.

Excessive vibration indicates possible fatigue risk.

7. Sliding Stability Check (Baseplate)

Since your footers are not permanently attached:

Method:

Apply lateral force equal to calculated max dynamic lateral load

Verify no baseplate sliding.

This is critical for competition compliance.

8. Electrical Continuity Test

Applicable To:

Emergency stop circuit

Sensor wiring

HMI wiring

Method:

Multimeter continuity check.

Confirm no broken wires or intermittent connections.

9. Functional Cycle Endurance Test

Applicable To:

Vehicle

Track

Lift mechanism

Method:

Run 25–50 continuous cycles.

Inspect for wear, cracking, or loosening afterward.

This is a fatigue screening test without damaging components.

12.4.1.1 Nondestructive testing includes components in a manner that will not impair the future performance or serviceability of these components

12.4.1.2 These tests will be used to verify the integrity of components to validate our MATLAB and Excel Calculations.

12.4.1.3 The ride system has been designed such that required major structural testing intervals shall not be more frequent than annually. Due to the prototype and competition-scale nature of the system, more frequent inspections are implemented during active operation to ensure continued safety and reliability.

Testing intervals are categorized as:

Pre-Operation (Before Each Use)

Daily / Operational Period

Periodic (Cycle-Based)

Annual Structural Review

2. Pre-Operation Testing (Before Each Operating Session)

Frequency: Prior to each operating period

Applies To:

3D printed track

3D printed vehicle

Copper support columns

Wooden baseplate

Lift hill

Sensors and control system

Emergency stop

Actions:

Visual inspection for cracks, deformation, or loosened fasteners

Verify support columns remain plumb

Confirm no baseplate shifting

Functional test of sensors

Emergency stop verification

One full operational cycle without passengers

3. Operational Monitoring

Frequency: Continuous observation during operation

Actions:

Observe vibration levels

Listen for abnormal mechanical noise

Monitor vehicle tracking and lift engagement

Verify no lateral displacement at footings

Testing shall be halted immediately if abnormal behavior is observed.

4. Short-Term Periodic Testing

Frequency: Every 50–100 operational cycles or every 8 operating hours (whichever occurs first)

Applies To:

Lift mechanism

Fastener connections

Track joints

Support column connections

Actions:

Fastener retention check

Track alignment verification

Deflection check at highest load point

Inspection of PLA components for fatigue cracking

5. Intermediate Periodic Testing

Frequency: Every 6 months or 500 operational cycles (whichever occurs first)

Actions:

Detailed structural inspection of copper columns

Inspection of plywood baseplate for warping or moisture damage

Acoustic tap testing of 3D printed components

Electrical continuity testing

Reverification of calculated load paths

6. Annual Structural Review (Maximum Required Interval)

Frequency: Annually (not more frequent than once per year unless modification occurs)

This satisfies the ASTM requirement that the initial design expect a testing interval no more frequent than annually.

Actions:

Full structural review against original design loads

Recalculation of governing load case if modifications occurred

Verification of factor of safety

Detailed inspection of all load-bearing members

Replacement of any component showing fatigue or degradation

Documentation update and sign-off

7. Post-Modification Testing

Frequency: Immediately following any structural, mechanical, or control modification

Actions:

Repeat erection acceptance testing

Conduct minimum three successful operational cycles

Verify emergency stop and sensor performance

Design Intent Statement

The ride has been designed such that structural components, including copper support columns, wooden baseplate, and 3D printed track sections, maintain adequate strength and stability under calculated maximum dynamic design loads. Major structural testing is required no more frequently than annually, in accordance with ASTM F1193, provided no modifications or abnormal loading events occur.

12.4.1.4 Components to be tested and the test requirements have been listed in sections 12.3.2.1-12.3.2.3. These requirements shall be met prior to operation.

12.4.1.5 A list of the components as well as their locations will be provided within the SolidWorks drawing package.

12.4.1.6 Components found to have relevant indications that do not meet the acceptance criteria shall be replaced or reconditioned in accordance with F1193.

12.4.1.7 Components found to have relevant indications that do not meet the acceptance criteria or have been reconditioned shall be further tested at the regular schedule in accordance with 12.4.1.3

12.4.1.8 If the manual does not contain testing recommendations shall either provide a component listing or statement that no NDT is recommended on the ride or device as per criteria outline of 12.4.1.2. When a manufacturer's list of statements is not available, it may be compiled by a registered professional engineer or engineering agency or by an individual qualified by training and experience to compile such a list or statement based upon the ride or device's specifications and history and using acceptable engineering practices.

18.1.5 Section 10 Information Requirements:

10.1 Information requirements are listed below:

10.2 A manufacturer-issued information plate currently is not a part of the scope of Joop Adventures, as our team must conserve space to be a part of the Portable Envelope.

10.3 The rider will be on the ride for a total of 11.65 seconds. For specified amounts of time per each section of track, refer to the MATLAB code.

10.4 Due to the range between 4-5 grams in the REC Prompt [1], Joop Adventures does not require a specified unloading and loading procedure as it relates to weight distribution.

10.5 Operational environmental limitations were evaluated as part of the ride analysis. The ride is designed and intended exclusively for indoor operation within a controlled

environment. As such, exposure to natural weather conditions is limited.

Wind: The ride is not designed for outdoor use. Wind loading was considered not applicable due to indoor installation. Operation is restricted to indoor environments where wind exposure is negligible.

Rain and Moisture: The ride is not rated for exposure to precipitation. Operation is prohibited in the presence of rain, standing water, or active water intrusion. Electrical and mechanical systems are not designed for wet-environment operation. Cleaning procedures require full drying prior to energizing the system.

Salt and Corrosive Environments: The ride is not intended for use in coastal or corrosive industrial environments. Salt exposure, chemical mist, or corrosive airborne contaminants are prohibited. Indoor storage in a dry, climate-controlled space is required to prevent corrosion of fasteners, electrical contacts, and mechanical components.

Extreme Heat: Operation is restricted to ambient temperatures that do not exceed manufacturer-rated limits for electronic components and PLA structural elements. Elevated temperatures approaching the glass transition temperature of PLA (~60°C / 140°F) are prohibited. The ride shall not be operated near heat sources or in direct sunlight conditions that could cause material softening or deformation.

Extreme Cold: Operation below freezing temperatures is discouraged due to potential embrittlement of polymer components and reduced battery/electrical performance. The ride shall be operated in environments maintaining typical indoor temperature ranges (approximately 10°C–30°C / 50°F–86°F).

Summary Statement: The ride is approved for operation only in indoor, dry, climate-controlled environments. Outdoor installation, precipitation exposure, high wind conditions, corrosive atmospheres, or temperature extremes beyond standard indoor conditions are prohibited.

10.6 Riders will be restricted to the specified dimensions within the REC Prompt [1] as shown in Figure 3.

10.7 Joop Adventures will operate under the requirements listed in the Prompt [1] including utilizing 1 standard 120VAC outlet for the ride and less than 10 Amps total. Joop Adventures will operate under these limits: 9.5 Amps and 24 volts.

10.8 The minimum horsepower for the ride to operate is 0.003866 hp. Our ride is designed to be able to sustain this speed.

10.9 Omitted as Joop Adventures does not incorporate any water features.

10.10 Maximum height: 475 mm, width: 750 mm, and length: 750 mm

10.11 Since the coaster is stationary, the static information matches the dynamics information. Refer to 10.10

10.12 height: 250 mm, width: 250 mm, length: 700 mm, weight: 3.6 kg

10.13 1. General

This Fastener Schedule is issued by the ride manufacturer to identify fastener type, material, grade (where applicable), installation method, placement, and torque requirements in accordance with applicable ride safety documentation requirements. The current ride design utilizes minimal mechanical fasteners, as the majority of components are integrated assemblies or pin-connected systems.

No threaded structural fasteners are utilized in the primary load-bearing systems of the ride. Therefore, torque specifications are not applicable to structural components.

2. Fastener Identification Table

Item No.: F-01

Location: Lift Hill Entrance – Infrared Break Beam Sensor Housing

Fastener Type: Smooth Shank Finish Nail

Quantity: 4

Material: Low-Carbon Steel

Function: Secures 3D-Printed Sensor Housing to Plywood Structure

Installation Method: Hammer-driven into plywood substrate

Torque Specification: Not Applicable

Item No.: F-02

Location: Track Segment Connections

Fastener Type: Integrated Polymer Pin / Press-Fit Connector

Quantity: Multiple (per track joint)

Material: PLA (3D-Printed Polymer)

Function: Alignment and Shear Transfer Between Track Segments

Installation Method: Manual Press-Fit Engagement

Torque Specification: Not Applicable

3. Fastener Design Considerations

The finish nails (F-01) are utilized solely to secure a non-load-bearing 3D-printed sensor housing. These fasteners are not structural and are not subject to significant tensile or shear loads beyond the negligible weight of the sensor assembly. As impact-driven fasteners, torque specifications are not applicable. Installation is performed in accordance with standard woodworking practice, ensuring full seating and flush installation.

Track segment connections (F-02) utilize integrated polymer pin connectors. These connections are non-threaded and function as alignment and shear transfer features. No torque application is required. The pins are not tension-critical fasteners and do not serve as primary structural retention devices.

4. Inspection Requirements

All fasteners are subject to pre-operation visual inspection to verify:

Full seating and proper engagement

Absence of withdrawal or loosening

No visible cracking of surrounding material

Proper alignment of track joints

Any damaged or compromised fasteners shall be replaced prior to operation.

5. Compliance Statement

Based on the limited use of non-structural fasteners and the absence of torque-controlled threaded structural bolts, the current fastener configuration is considered appropriate for the scale and loading conditions of the ride. Torque specifications are not required due to the absence of threaded structural fasteners.

10.14 the Load per column is 0.287 N, the vertical load per column is 0.0260 N, lateral load is 0.2482 N, the total dynamic load with the impact factor is 11.490 N. Further information can be found in the MATLAB as well as the appendix for the ANSYS calculations (*Appendix Figures 17-22*). Results from mesh sizes were compared with the smallest mesh size possible to ensure that all results were within 5% error. This data can further be explored in the Excel calculations. The support columns simulated had a 2 N force applied to the top to simulate the 105 gram ride vehicle weight as well as the force of gravity for the longest and the shortest support tubes in the assembly.

10.15 The following calculations were conducted to confirm that Joop Adventures would operate appropriately.

10.15.1 Maximum static loads for each of the support columns were conducted in ANSYS (*Appendix Figures 17-22*). Additional buckling and tipping analyses were conducted in MATLAB.

10.15.2 The maximum dynamic design load for each support column (structural connection point) was calculated using the governing lateral and vertical load conditions, including a conservative dynamic impact factor of 1.5 to account for acceleration and transient effects. With 40 evenly spaced support columns, the resulting maximum load per footing is 0.287 N, with vertical and lateral components of 0.026 N and 0.248 N, respectively. These loads define the design requirements for each footing and structural connection. These calculations can further be explored in the MATLAB code.

10.15.3 All other structural interface design specifications can be found within the MATLAB code.

COATINGS

19.1 Material coatings are applied to the station structure, baseplate, laboratory enclosure, and HMI housing assemblies. These components are constructed from pine wood and plywood and are finished with paint coatings serving as protective surface barriers.

Potential degradation due to environmental conditions was evaluated in the ride analysis. The coatings function to reduce moisture absorption, surface wear, and biological growth associated with untreated wood materials. The ride is intended for indoor operation in a controlled environment, thereby limiting exposure to ultraviolet radiation, precipitation, and extreme temperature variation. As a result, environmental degradation mechanisms are minimized.

The coatings are non-structural and do not contribute to primary load-bearing capacity. Therefore, minor surface wear or cosmetic degradation would not compromise structural integrity. Coated components are subject to routine visual inspection to identify any deterioration requiring maintenance or reapplication.

Based on these considerations, the design accounts for potential degradation associated with environmental conditions.

19.2 Material coatings were selected and applied in accordance with the respective manufacturer's instructions. Surface preparation, application method, drying/curing time, and ventilation requirements were followed to ensure proper adhesion and performance.

Associated Safety Data Sheets (SDS) were reviewed to confirm safe handling procedures, required personal protective equipment during application, ventilation requirements, storage guidance, and chemical compatibility with plywood substrates.

The applied coatings do not introduce additional hazards to patrons or operators once cured. Electrical and mechanical systems were protected from coating overspray and chemical exposure during application.

Based on adherence to manufacturer instructions and SDS guidance, the coating application is considered compliant with Section 19.2 requirements.

FALL PROTECTION

20.1 Fall protection requirements were evaluated as part of the ride safety analysis. The maximum elevation of the ride structure is 475 mm (18.7 in) above the surrounding floor surface. This height does not constitute an elevated walking-working surface requiring fall protection measures under typical safety thresholds.

All patron loading and unloading occurs at ground level, and no elevated platforms, maintenance walkways, or operator stations are incorporated into the design. As a result, fall arrest systems, guardrail systems for fall protection, and personal protective equipment were determined to be not applicable to this ride configuration.

Based on the limited maximum height and ground-level configuration, fall protection measures are not required.

SANITATION/DISINFECTION

21.1 Sanitation and disinfection factors were evaluated and incorporated into the overall ride analysis. The ride incorporates 3D-printed polymer track components and plywood structural elements within the station, lab interface, and baseplate assemblies. The material composition and surface configuration

of these components were assessed to evaluate their susceptibility to moisture absorption, contamination, and potential biological growth.

The 3D-printed polymer track surfaces were considered low-porosity but contain layer lines inherent to additive manufacturing; therefore, cleaning procedures were selected to ensure adequate surface coverage without causing material degradation. Plywood components, being porous wood-based materials, were evaluated for moisture exposure and were either sealed or designated as limited-contact surfaces to reduce biological growth potential. High-touch areas, including restraints, gates, and station interfaces, were specifically identified in the analysis.

The nature and frequency of patron contact were reviewed to determine disinfection frequency. A routine cleaning and disinfection schedule proportional to operational hours was established. Selected disinfectant agents were reviewed in accordance with their associated Safety Data Sheets (SDS) to confirm compatibility with polymer and wood materials and to ensure no adverse impact on structural integrity, mechanical systems, electrical systems, or control systems.

These sanitation, surface configuration, biological growth potential, and disinfectant compatibility considerations were formally incorporated into the overall ride safety and operational analysis.

KEYWORDS

22.1 Acceleration Limits; Adafruit Infrared Break Beam Sensor; Amusement Ride or Devices; Anti-Rollback Devices; Automated Forced Stop Procedure; Chain; Clearance Envelope; C-More CM5-T7W Human-Machine Interface; Containment; Control Systems; Design; DROK 48V Adjustable Power Supply; DROK DC Buck Converter; Electrical; Elegoo Arduino Uno; Emergency Safety Device; Fasteners; Fatigue; Fencing; FS90R Continuous Servo Motor; Gates; Guardrails; Impact Factor; LAY7 Emergency Stop Push Button; Loads and Strengths; Mechanical Systems; MG90S Positional Servo Motor; Operational Hours; Restraints; Ride Analysis; Ride Vehicle; Sprocket; Station; Track; Xilinx Basys 3.

[1] Ride Engineering Competition, "Ride Engineering Competition 2026 Prompt: Thrills on the Go: Portable Ride Engineering Challenge." Available: <https://www.rideengineeringcompetition.org/>.

[2] Ride Engineering Competition, "Ride Engineering Competition ASTM Adaptation Document 2026." Available: <https://www.rideengineeringcompetition.org/>

APPENDIX

Eligibility Criterion	Rationale/Hazard	Enforcement	Residual Risk
Minimum rider height of 22.2 mm	Prevents the ride harness from improperly securing the rider - too small	Appropriate signage Operator enforcement	Low
Maximum rider height of 25.2 mm	Prevents the ride harness from improperly securing the rider - too large	Appropriate signage Operator enforcement	Low
Riders must sit upright throughout the duration of the ride	Risk of contact with ride structure or elements	Appropriate signage Operator enforcement	Low
Riders must keep appendages in ride vehicle at all times	Risk of contact with ride structure or elements	Appropriate signage Operator enforcement	Low
Riders with medical conditions that may be aggravated during the ride are not permitted to ride	Minimizes the risk of injury	Appropriate signage Operator enforcement	Low

Figure 4: Patron Suitability Assessment Table

Restraint System Component	Hazard Addressed	Failure Mode	Mitigation	Residual Risk
Ratcheting Bottom Bar Restraint	Rider ejection	Lap bar improperly or not fully engaged Lap bar disengaged in the middle of the ride	Mechanical ratcheting and locking mechanism Operator verification	Low
Secondary Top Bar Restraint	Rider ejection	Lap bar improperly or not fully engaged lap bar disengaged in the middle of the ride	Latching position Operator verification	Low
Ratcheting Bottom Bar Restraint Ratchet	Loss of restraint integrity	Lap bar ratcheting mechanism wearing out due to repeated use	Operator visual inspection and routine maintenance checks	Low
Secondary Top Bar Restraint Hinge	Loss of restraint integrity	Lap bar hinge wearing out due to repeated use	Operator visual inspection and routine maintenance checks	Low
Ride vehicle sidewalls	Unplanned rider movement during ride	Loss of sidewall integrity	Operator visual inspection and routine maintenance checks	Low

Figure 5: Patron Restraint and Containment Table

Ride Segment / Location	Clearance Concern	Applicable Clearance Requirement	Residual Risk
Station	Rider makes contact with station structure	Minimum of 18.95 mm clearance is maintained around the ride vehicle (measured from the edge of the outer ball bearing on the wheel bogie to the friction wheel servo motor protruding from the side of the station)	Low
Rider Gates	Rider trapped in station or harmed due to pinch point	Maintain a minimum of 1.4 mm clearance between the edge of the maximum rider width (20.2 mm) and the edge of the gate.	Low
Ride Vehicle Harness	Harness serves as a pinch point for rider	Primary ride harness is designed to rest on the rider's waist, preventing them from being ejected from the vehicle	Low
Ball Bearings on Ride Vehicle	Outer ball bearings on chassis make contact with scenic element or support structure	Maintain a minimum of 13 mm clearance between any element and the edge of the ball bearing to ensure that the rider reach envelope (10 mm) and additional clearance envelope outside their reach (3 mm) are maintained	Low
Lifthill ascent	Rider comes in contact with support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low
Lifthill descent	Rider comes in contact with support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low
Straight track sections	Rider comes in contact with support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low

Figure 6: Patron Clearance Envelope 1

Ride Segment / Location	Clearance Concern	Applicable Clearance Requirement	Residual Risk
Track turn sections	support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low
Helix track section	Rider comes in contact with support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low
45-degree banked turn track section	Rider comes in contact with support structure, electrical component, or scenic component	Patron envelope (13 mm) maintained	Low

Figure 7: Patron Clearance Envelope 2

Activity	Hazard	Hazardous Situation	Mitigation	Residual Risk
Entering/exiting vehicle	Fall or slip	Rider falls or slips when rider enters or exits the vehicle	Leveled loading platform	Low
Improper restraint securing	Improper seating position	Rider is improperly restrained in ride vehicle	Operator visual check Two forms of rider restraints	Low
Dispatch of ride vehicle	Ride vehicle exits station before allotted time	Break beam sensors detect where both train cars are located on the track at an instant in time to prevent collisions	Autonomous system with alerts if the ride vehicles are outside the allotted time frame	Low
Emergency Stop	Immediate deceleration	Rider experiences extreme G force	Low operating speeds in station when vehicle is passing over friction wheels	Low
Rider unloading	Contact with vehicle, station, or support columns	Rider contacts vehicle, station or support columns in a way that causes discomfort, pinches them or restricts their motion that would otherwise normally be prohibited	Low operating speeds in station	Low

Figure 7: Load/Unload Activities

Component	Failure Mode	Cause of failure	Effect of Failure	Mitigation	Residual Risk
Lift hill chain	chain breaks	material fatigue	Vehicle stops on lift hill or falls backwards	Break beam sensors monitor ride spacing to prevent rides from coming too close to each other to prevent a collision on the track	Low
Lift hill motor	Motor failure	Over or under current	Vehicle stops on lift hill or falls backwards	Sprockets used to propel the car up the lift hill are equipped with the ability to prevent rollback, by only rotating forwards	Low
Break beam sensor	Sensor fails to detect vehicle	Improper wiring electronic failure	Faulty emergency stop or vehicles crash	Visual operator checks and regular monitoring on Human-Machine Interface	Low
Magnetic brakes	Brake fails to slow vehicle down enough	Improper magnet placement underneath ride vehicle chassis	Riders experience extreme deceleration in the station by hitting the friction wheels	Test to optimize for proper tolerance of magnet underneath the body of the vehicle	Low
Friction Wheels	Servo motors do not rotate	Electronic failure	Riders are stuck on the track either before or inside the station	Test the servo motors are rotating properly before operation Additional visual checks throughout the operating day	Low

Figure 8: Failure Analysis 1

Component	Failure Mode	Cause of failure	Effect of Failure	Mitigation	Residual Risk
Friction Wheels	Servo motors over-rotate	Electronic failure	Ride goes through station without stopping, potentially leading to ride vehicle collision	Test the servo motors are rotating properly before operation Additional visual checks throughout the operating day	Low
Primary waist lap restraint	Lap bar does not lock	Material fatigue	Rider is not secured into the ride vehicle throughout the duration of the ride	Optimize lap restraint strength by orienting the layer lines perpendicular to shear force	Low
Secondary top restraint	Lap bar does not latch	Material fatigue	Rider is not secured into the ride vehicle throughout the duration of the ride	Optimize lap restraint strength by orienting the layer lines perpendicular to shear force	Low
Ride Vehicle frame and chassis	Structural integrity	Material fatigue	Ride vehicle has the potential to break midway through the ride	Operator inspection before each launch Routine maintenance is performed	Low
Vehicle sidewalls	Structural integrity	Material fatigue	Rider is not secured from lateral forces throughout the duration of the ride	Operator inspection before each launch Routine maintenance is performed	Low
Emergency stop	Fails to initiate	Mechanical or electrical failure	Electrical components of the ride do not shut off	Test the emergency stop button Routine maintenance to prevent material fatigue Multiple stopping places along the track	Low

Figure 9: Failure Analysis Part 2

Component	Failure Mode	Cause of Failure	Effect of Failure	Mitigation	Residual Risk
Human-Machine Interface	Power Loss	Material fatigue	Software issue Power outage	Include several locations where the ride vehicle can stop to ensure that the patrons will come to a safe stop	Low
Buck Converter	Electrical Failure	Motor failure	Electrical components of the ride will not function properly	Test the ride vehicle to ensure that it will safely come to a stop in any orientation Test code to ensure bugs are not present	Low
Track Supports	Deformation	Material fatigue	Visual inspection ANSYS simulation to ensure that the support columns will not deform	Secure the copper support tubing into the wood bases Secure the copper tubing into the track	Low
Ride Vehicle Ball Bearings	Wheels seize up	Material fatigue	Ride vehicle gets stuck on track	Sufficiently lubricate ball bearings prior to operation Operator routine checks	Low

Figure 10: Failure Analysis 3

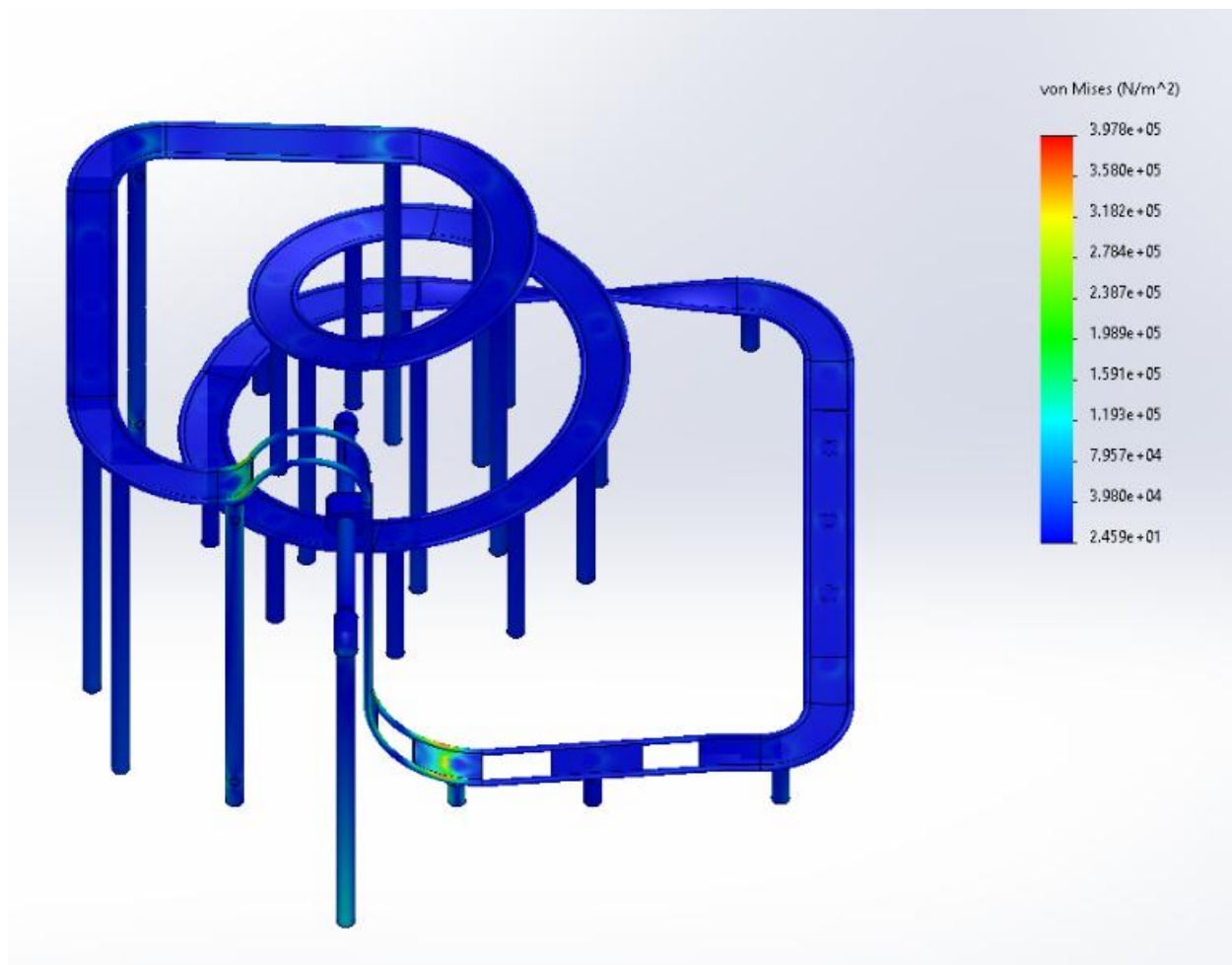


Figure 11: Stress distribution along the roller coaster track under maximum load (2 N) and including gravity 1

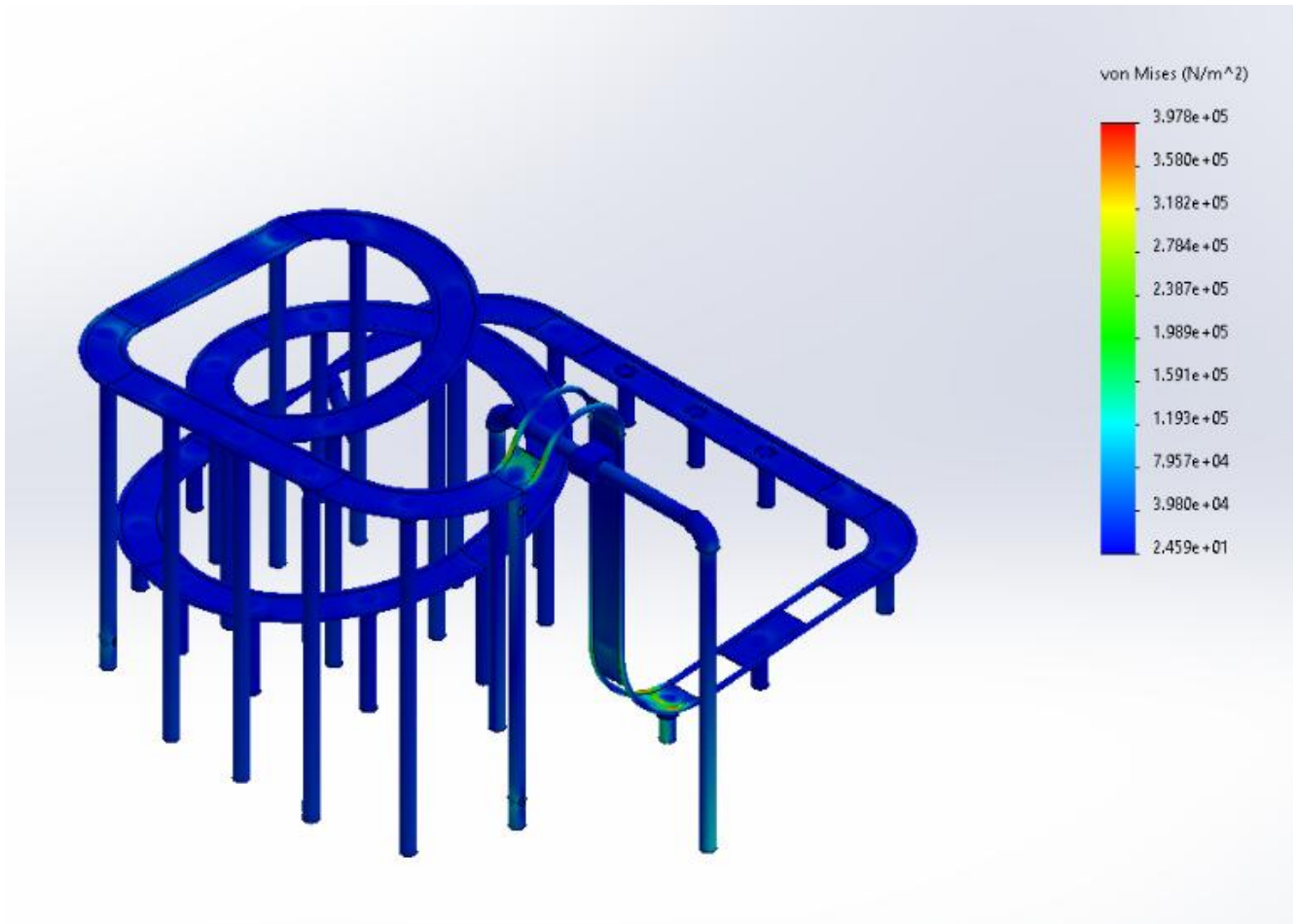


Figure 12: Stress distribution along the roller coaster track under maximum load (2 N) and including gravity 2

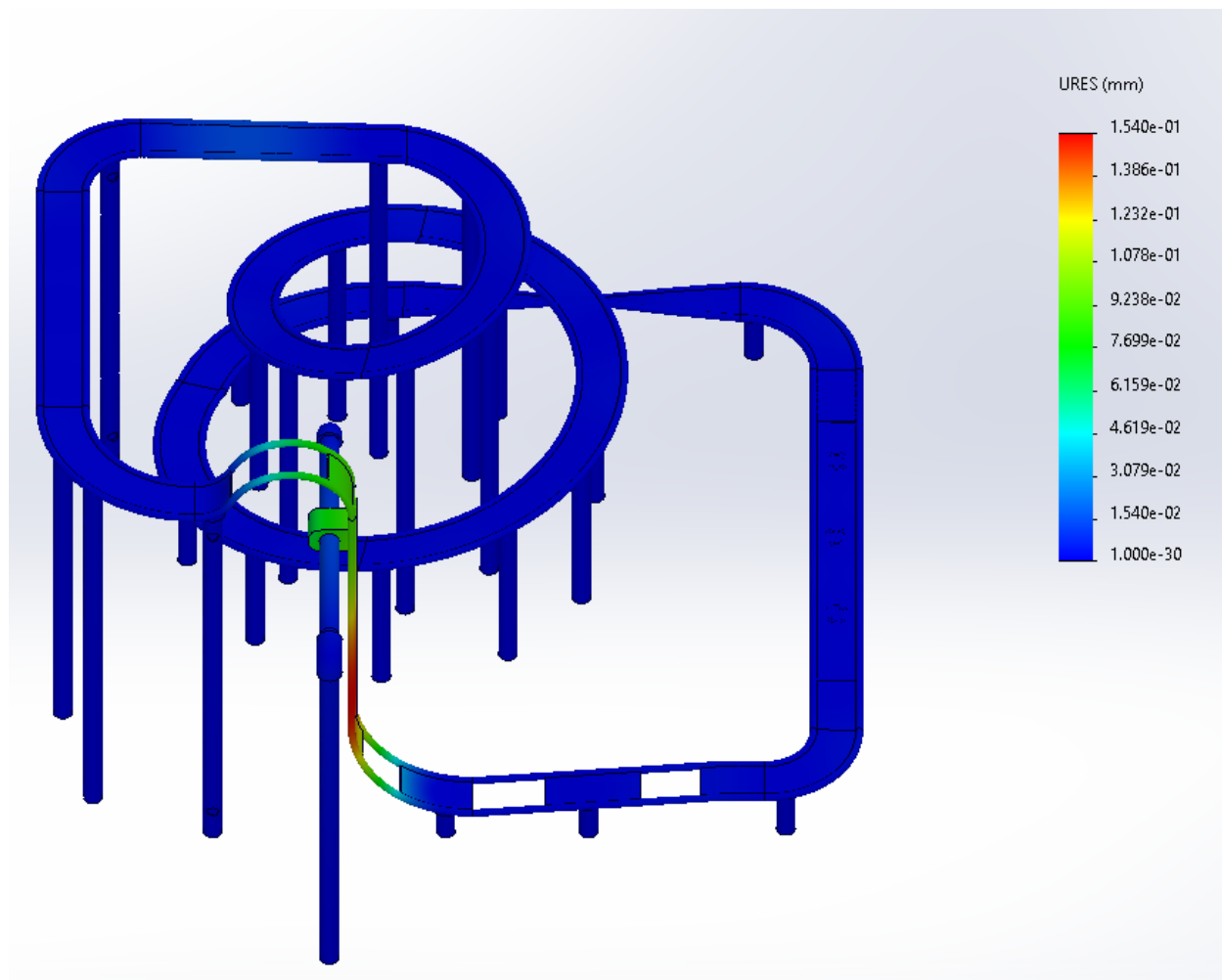


Figure 13: Displacement of Track 1

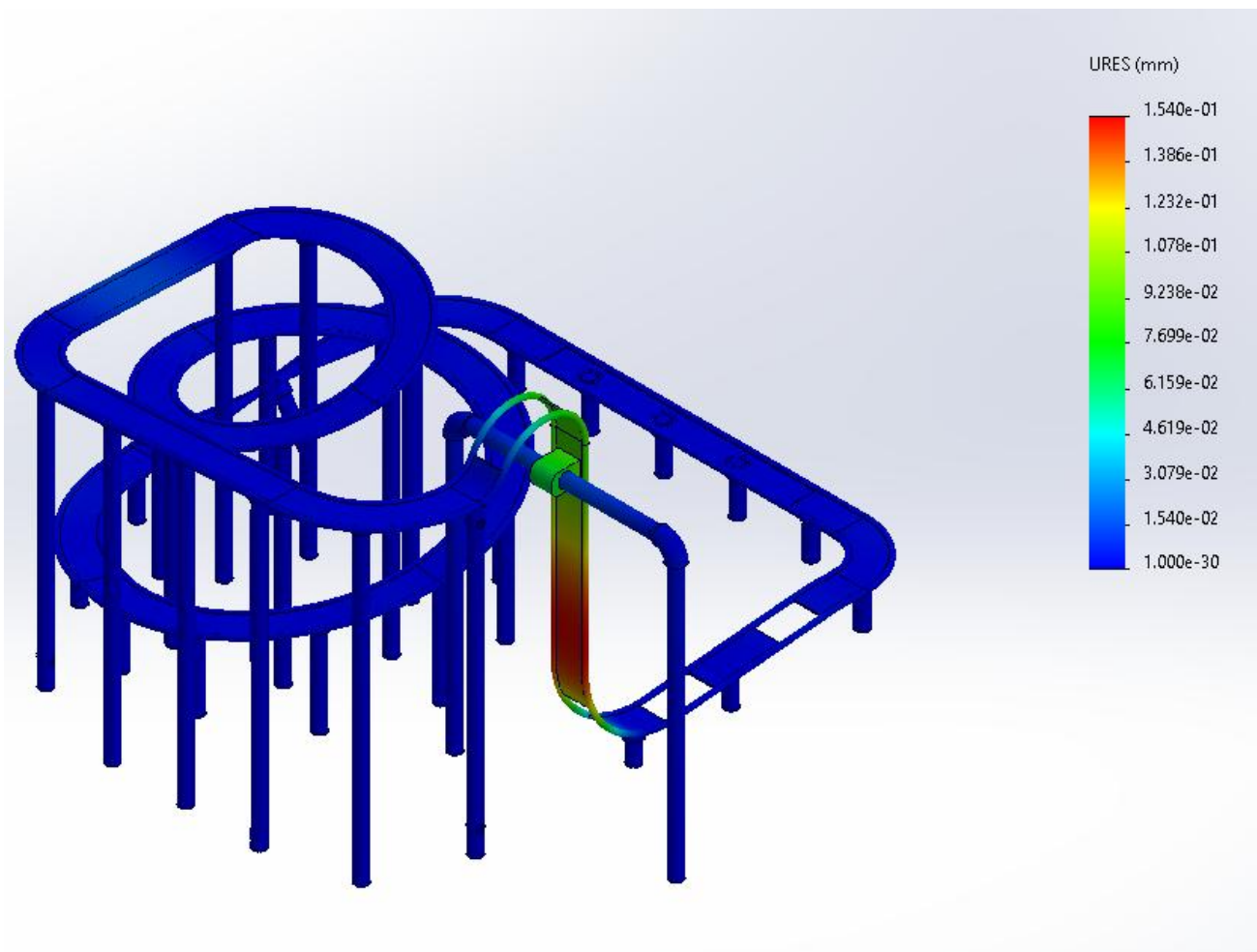


Figure 14: Displacement of Track 2

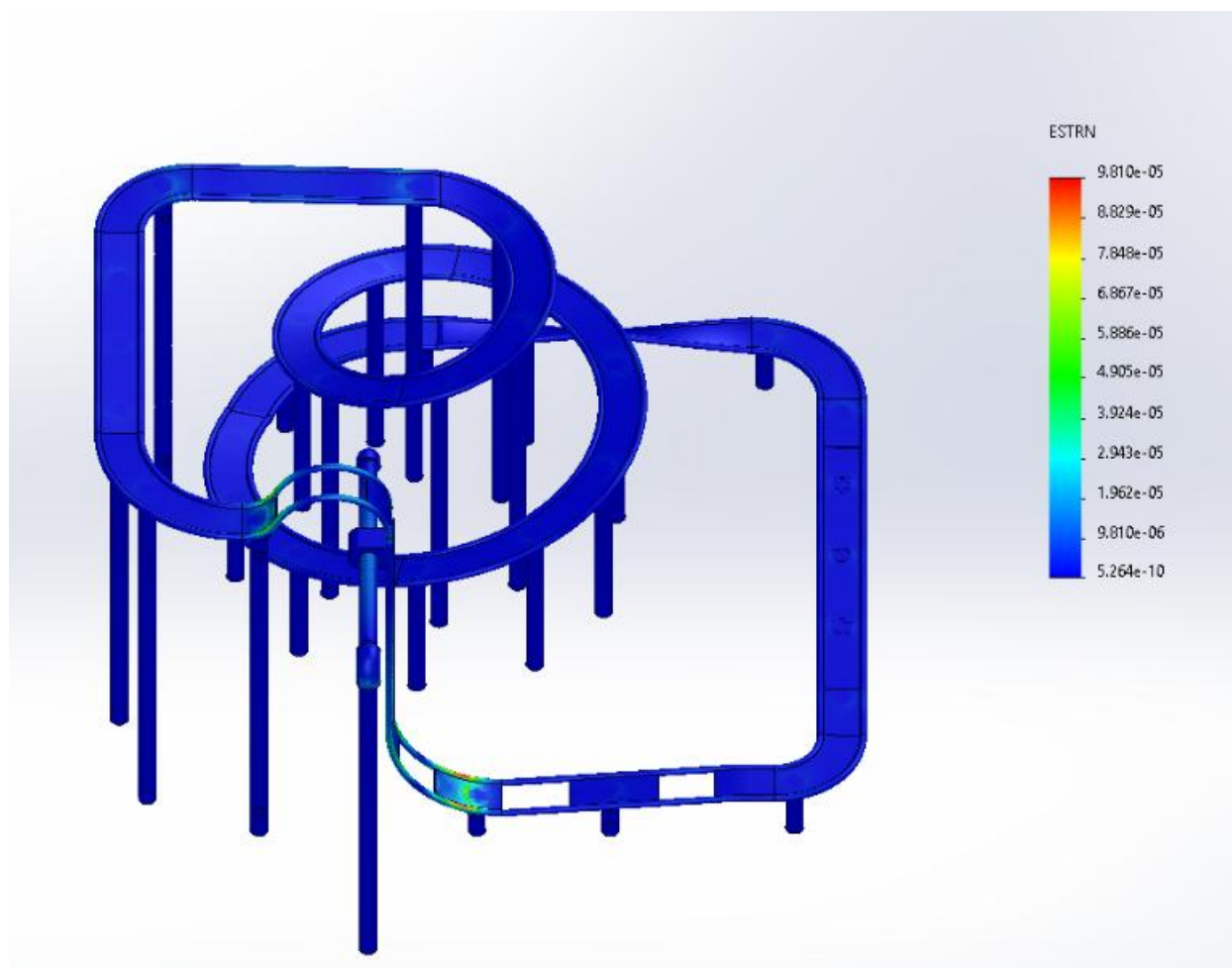


Figure 15: Strain 1

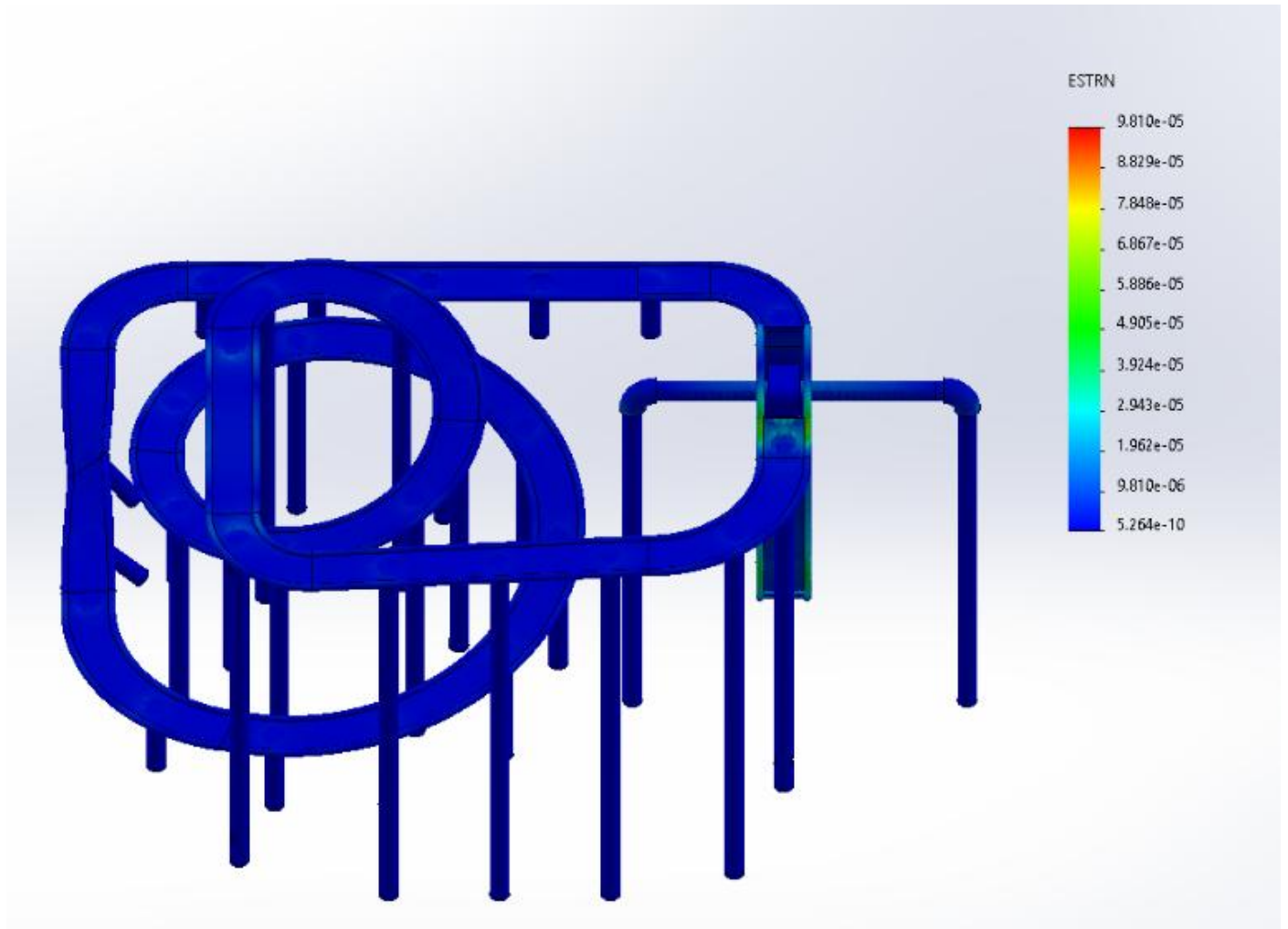


Figure 16: Strain 2

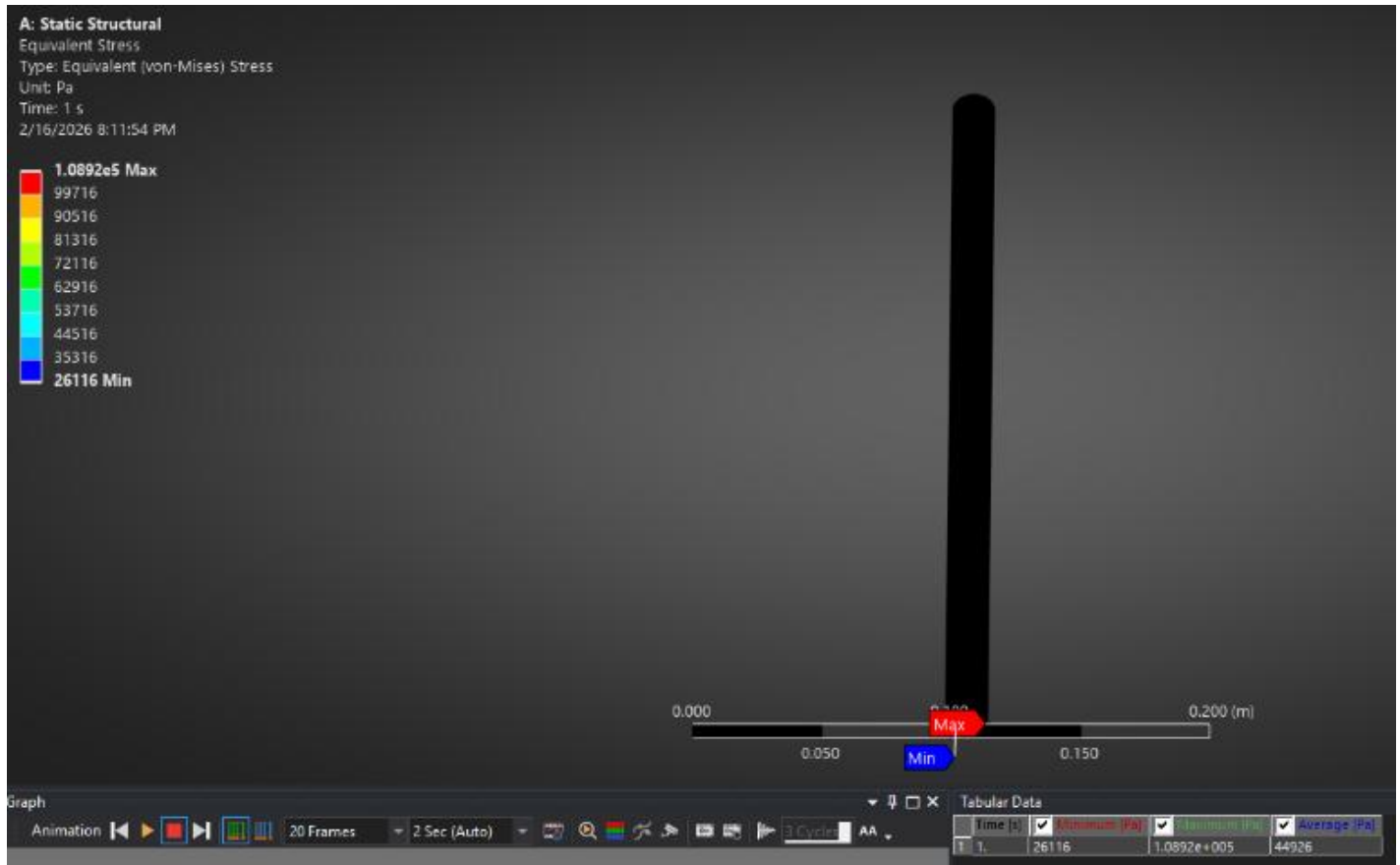


Figure 17: Stress distribution of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm). It appears black due to small mesh size.

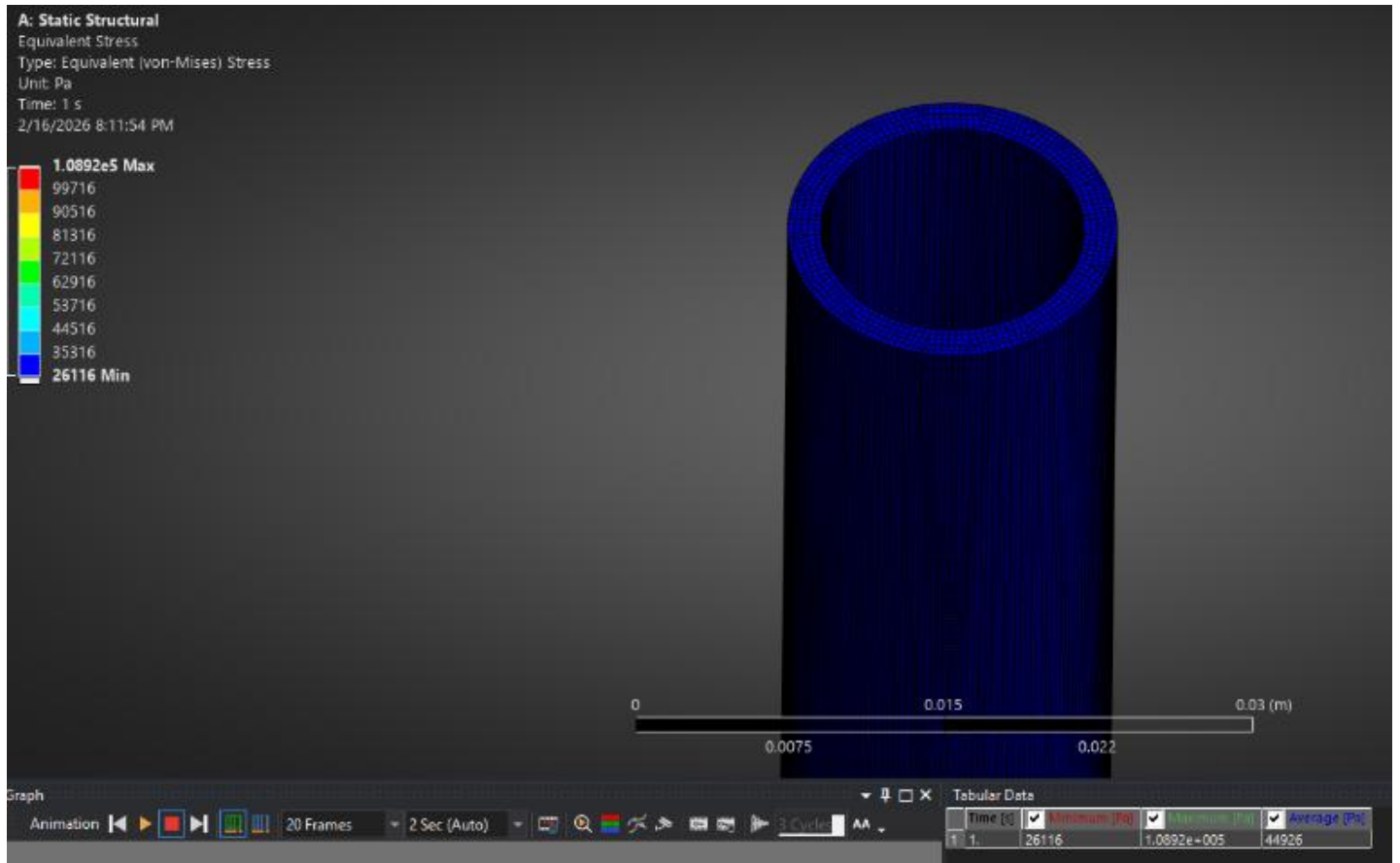


Figure 18: Top of Stress distribution of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm).

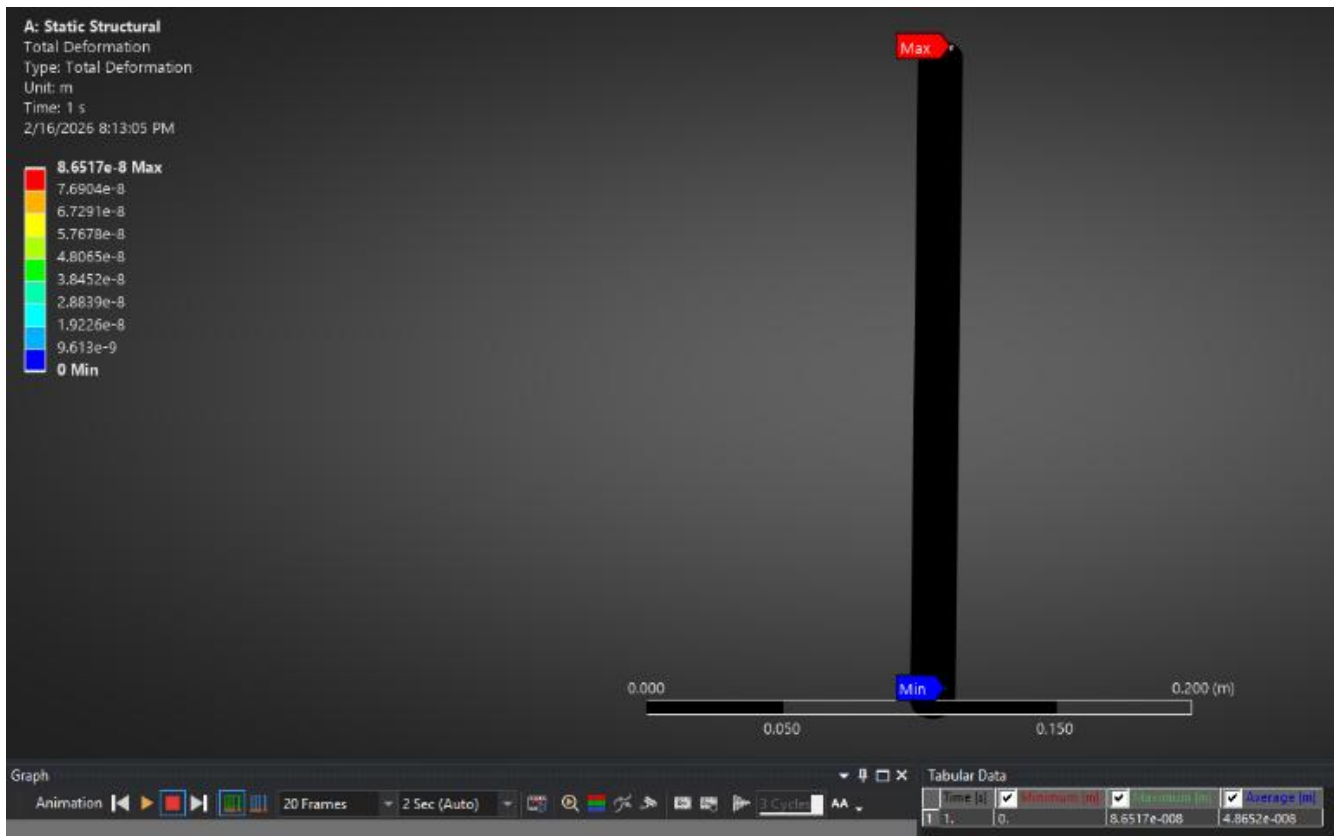


Figure 19: Deformation of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm). It appears black due to small mesh size.

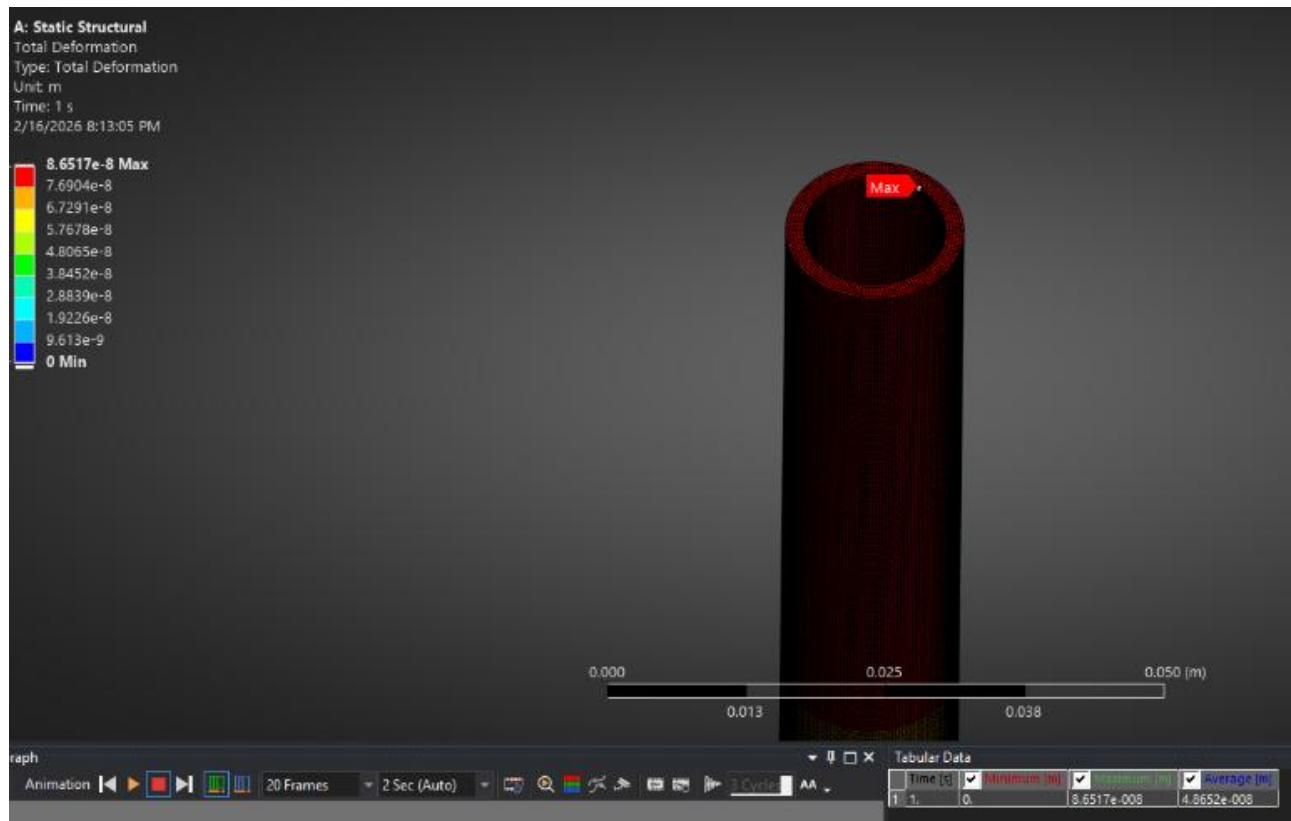


Figure 20: Top of Tube. Deformation of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm)

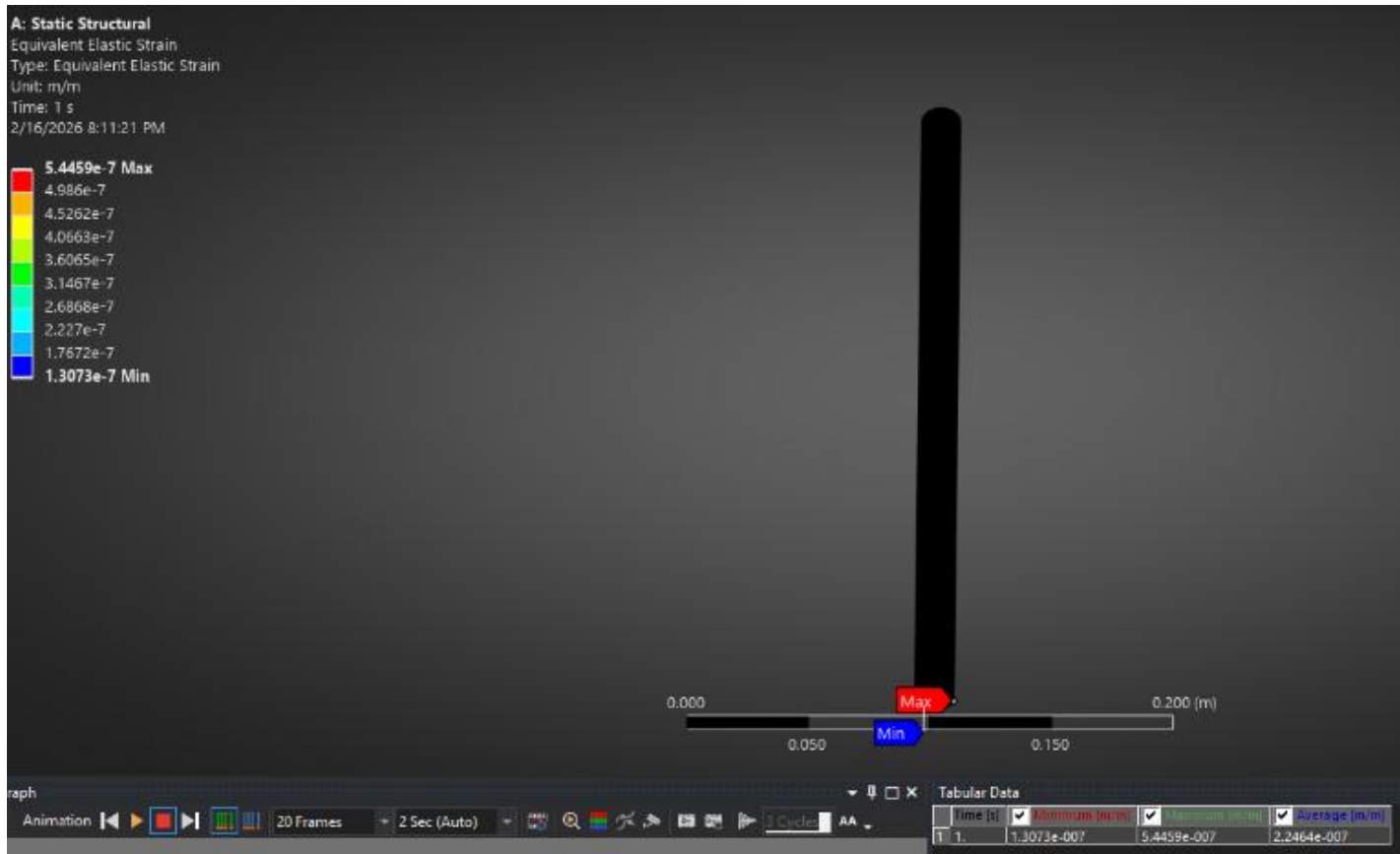


Figure 21: Strain distribution of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm). It appears black due to small mesh size.

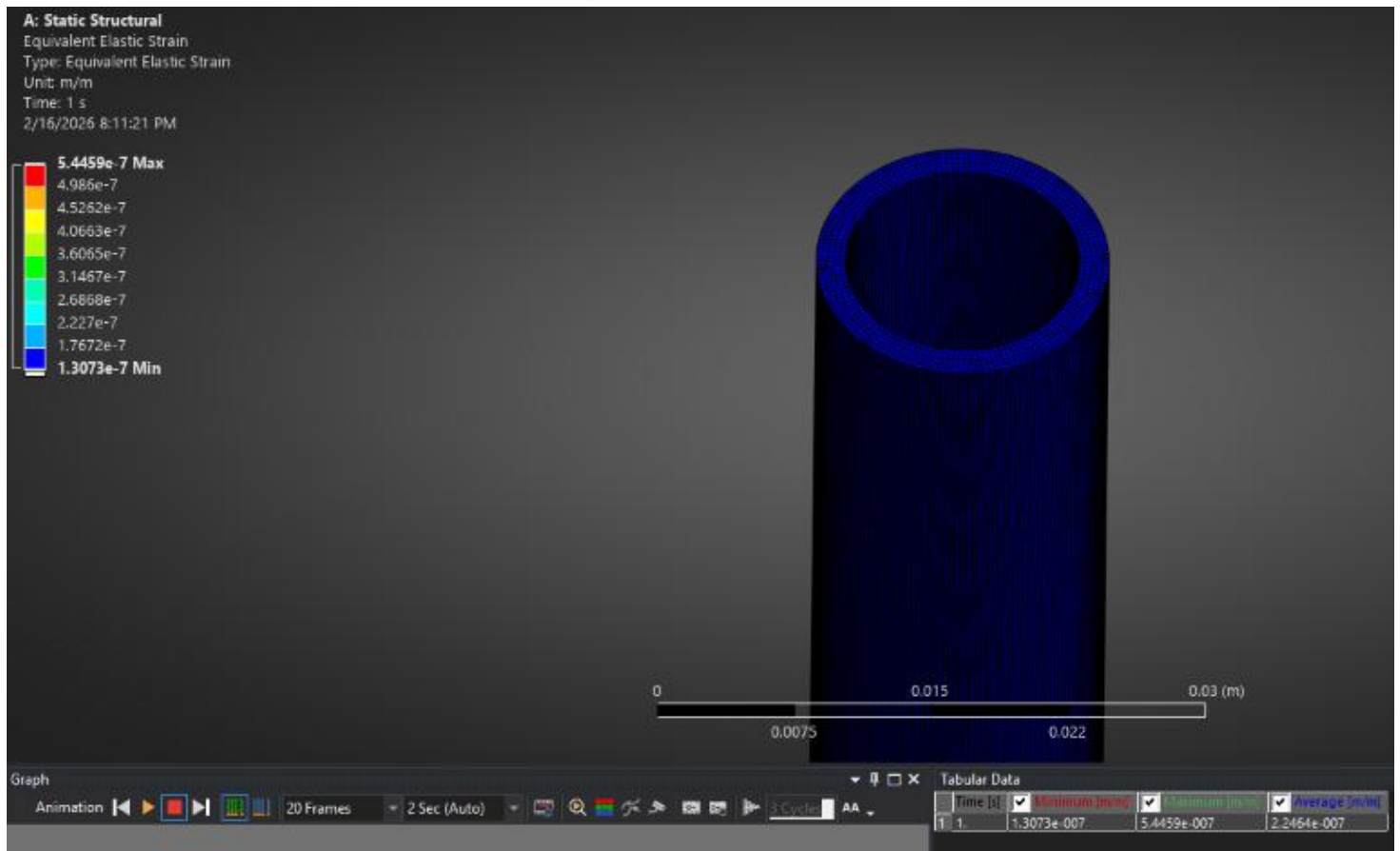


Figure 22: Strain Distribution of longest tube with gravity and applied load (1.97 N) to simulate track and car mass. Longest copper tube within assembly (385 mm).

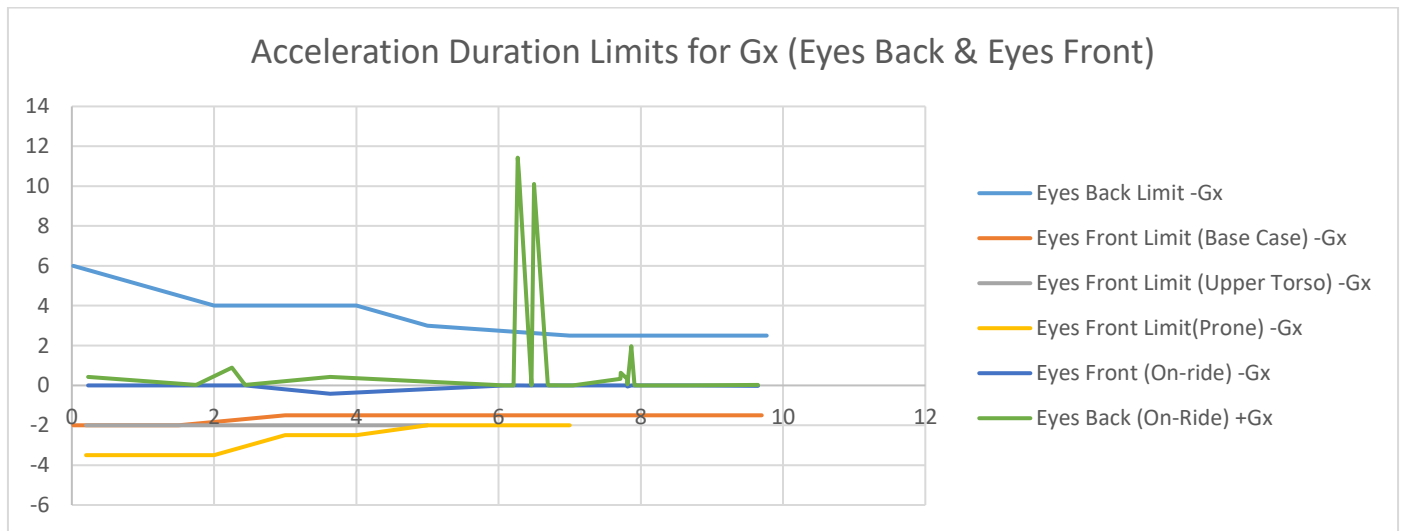


Figure 23: Acceleration Duration Limits for Gx (Eyes Back and Eyes Front)

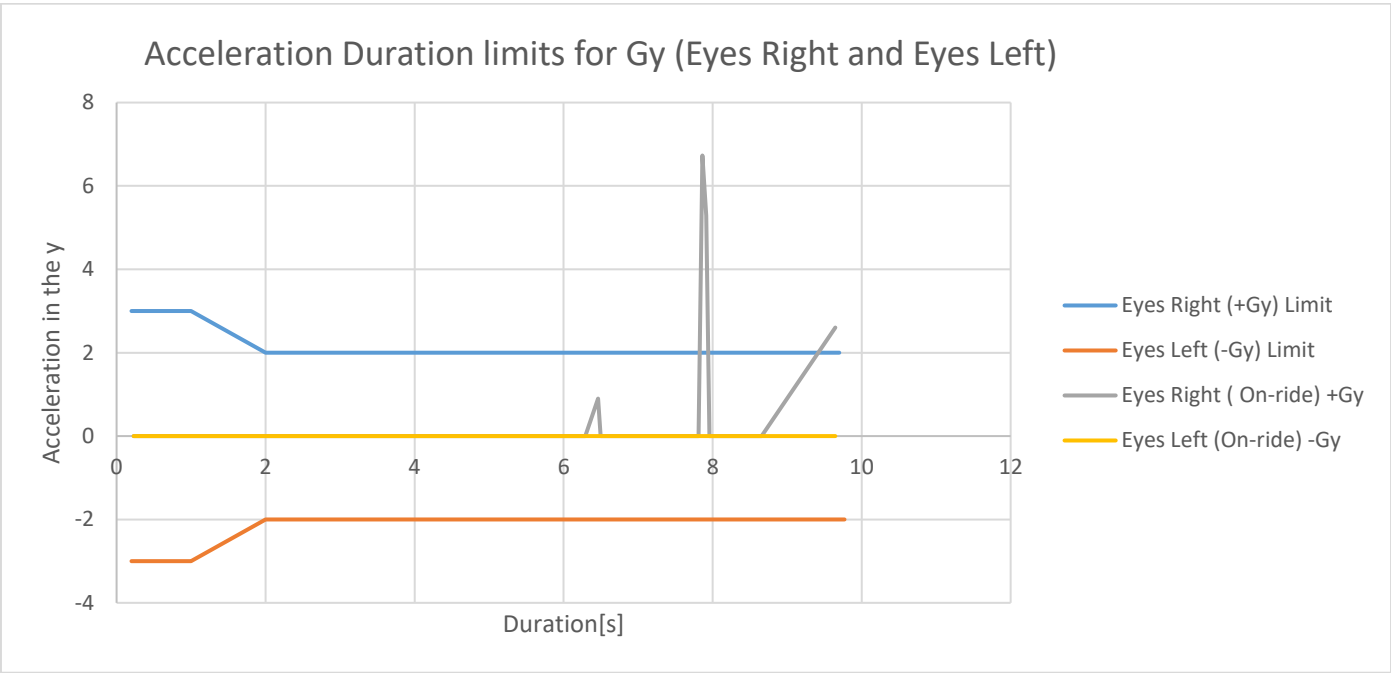


Figure 24: Acceleration Duration Limits for Gy (Eyes Right and Eyes Left)

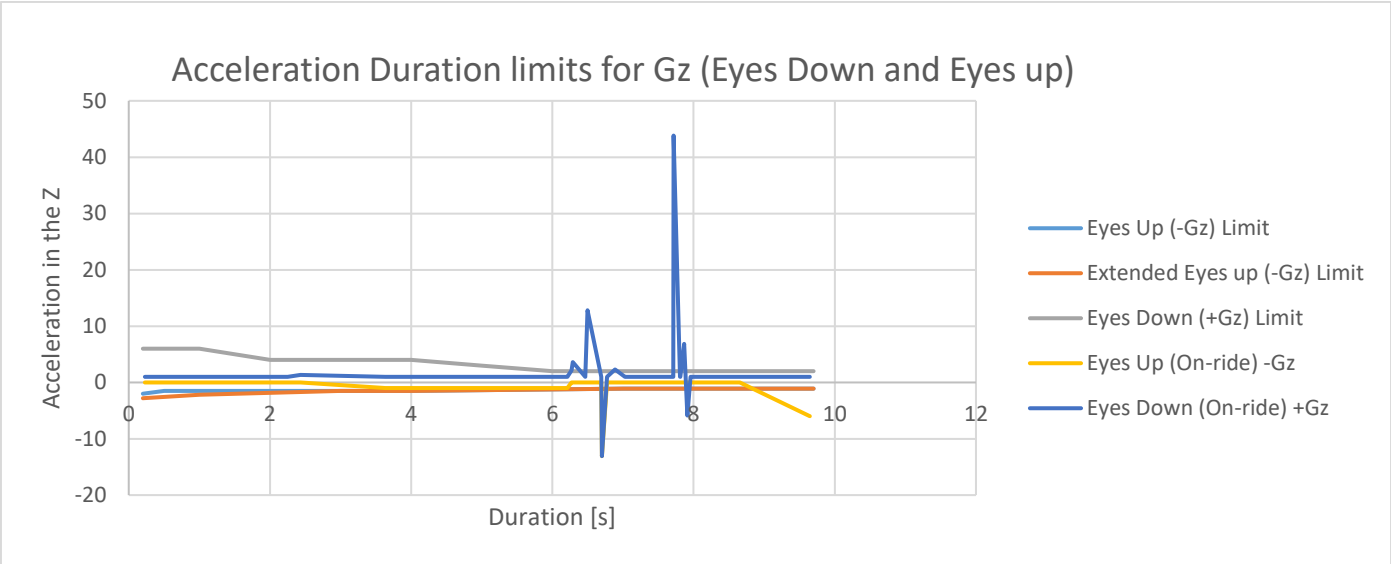


Figure 25: Acceleration Limits for Gz (Eyes Down and Eyes Up)

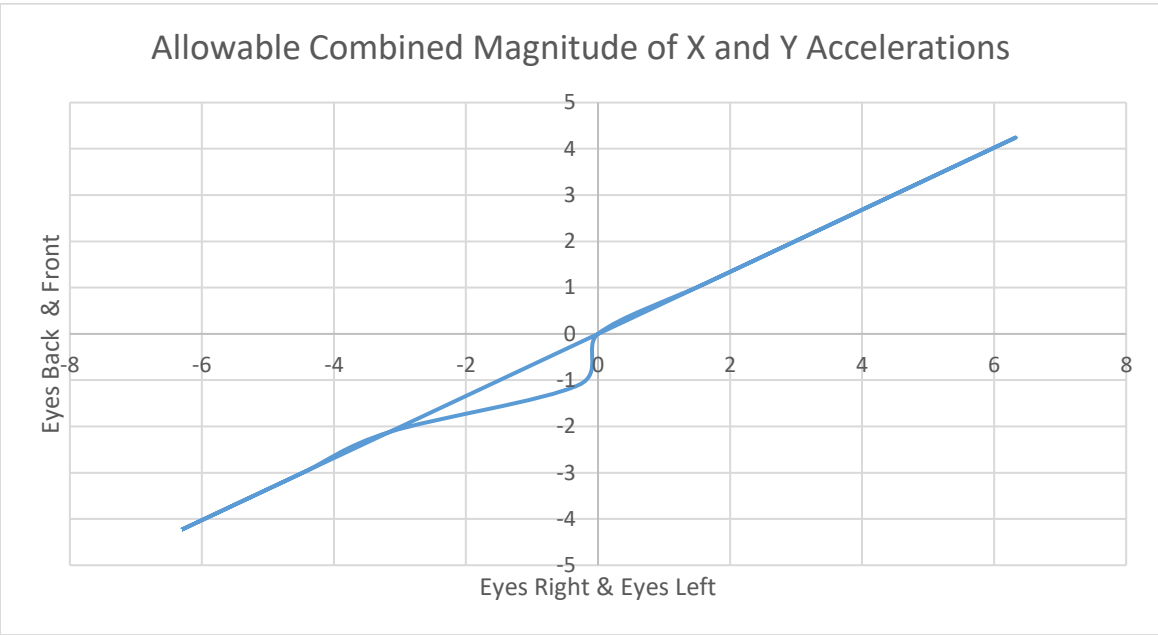


Figure 26: Allowable Combined Magnitude of X and Y Accelerations

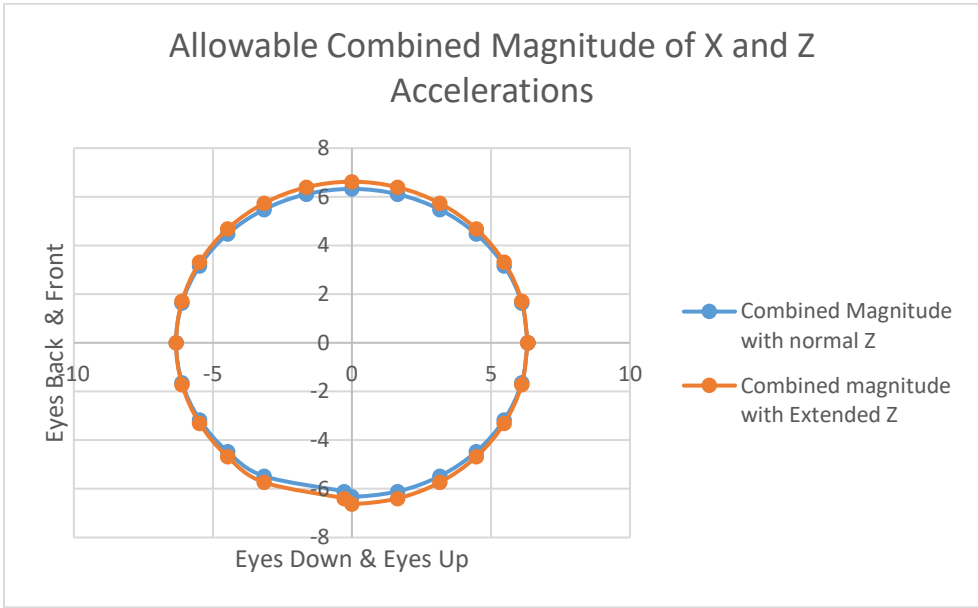


Figure 27: Allowable Combined Magnitude of X and Z Accelerations

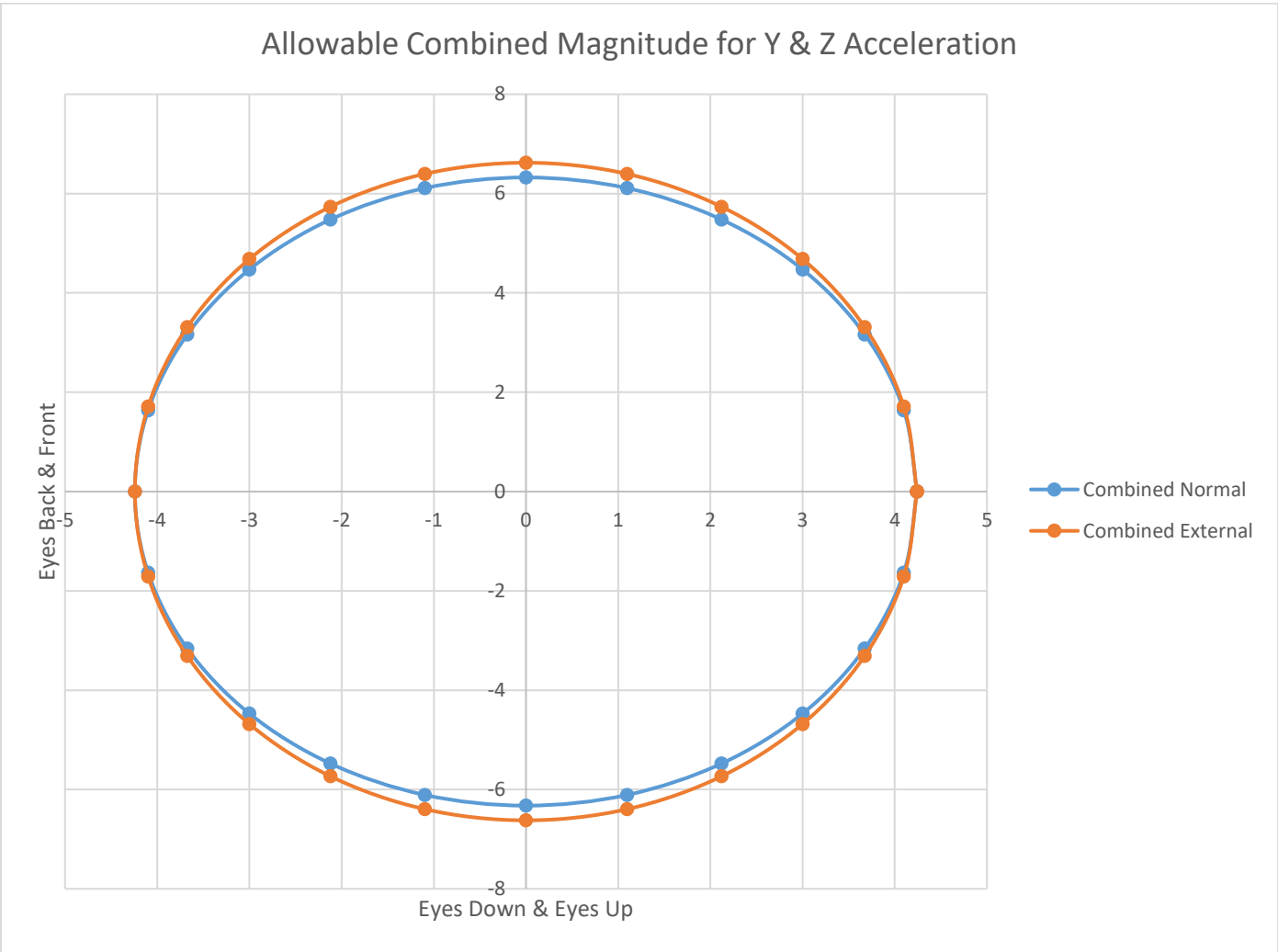


Figure 28: Allowable Combined Magnitude for Y & Z Acceleration

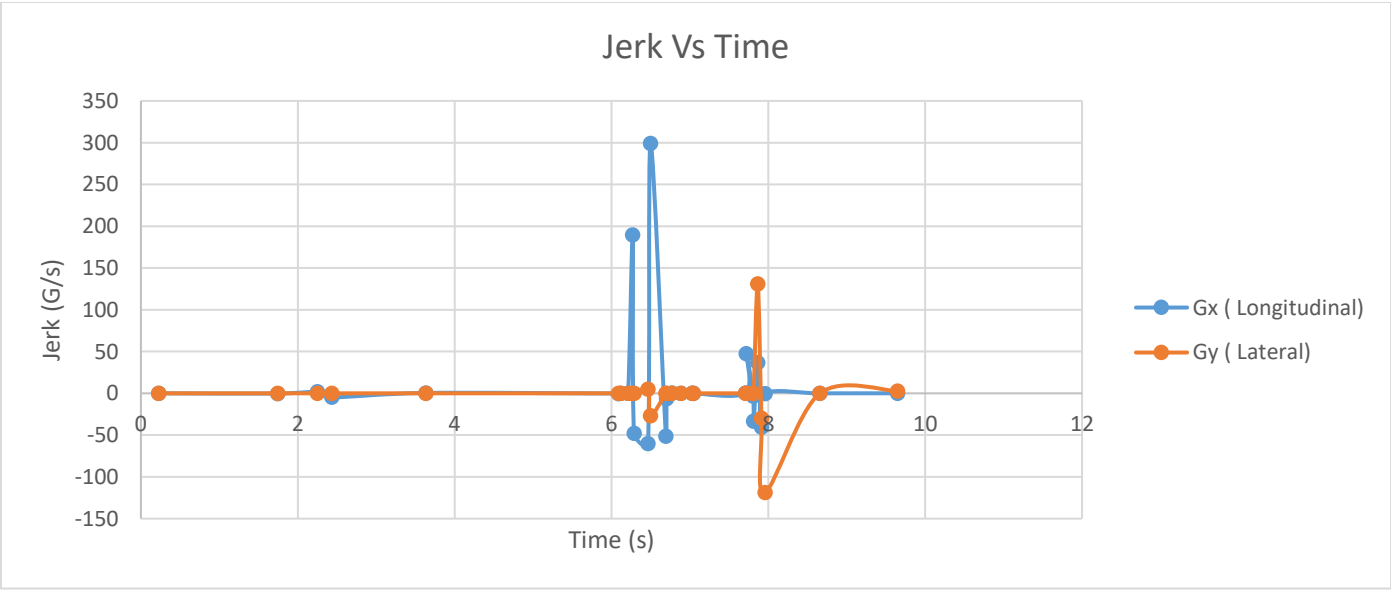


Figure 29: Jerk vs Time

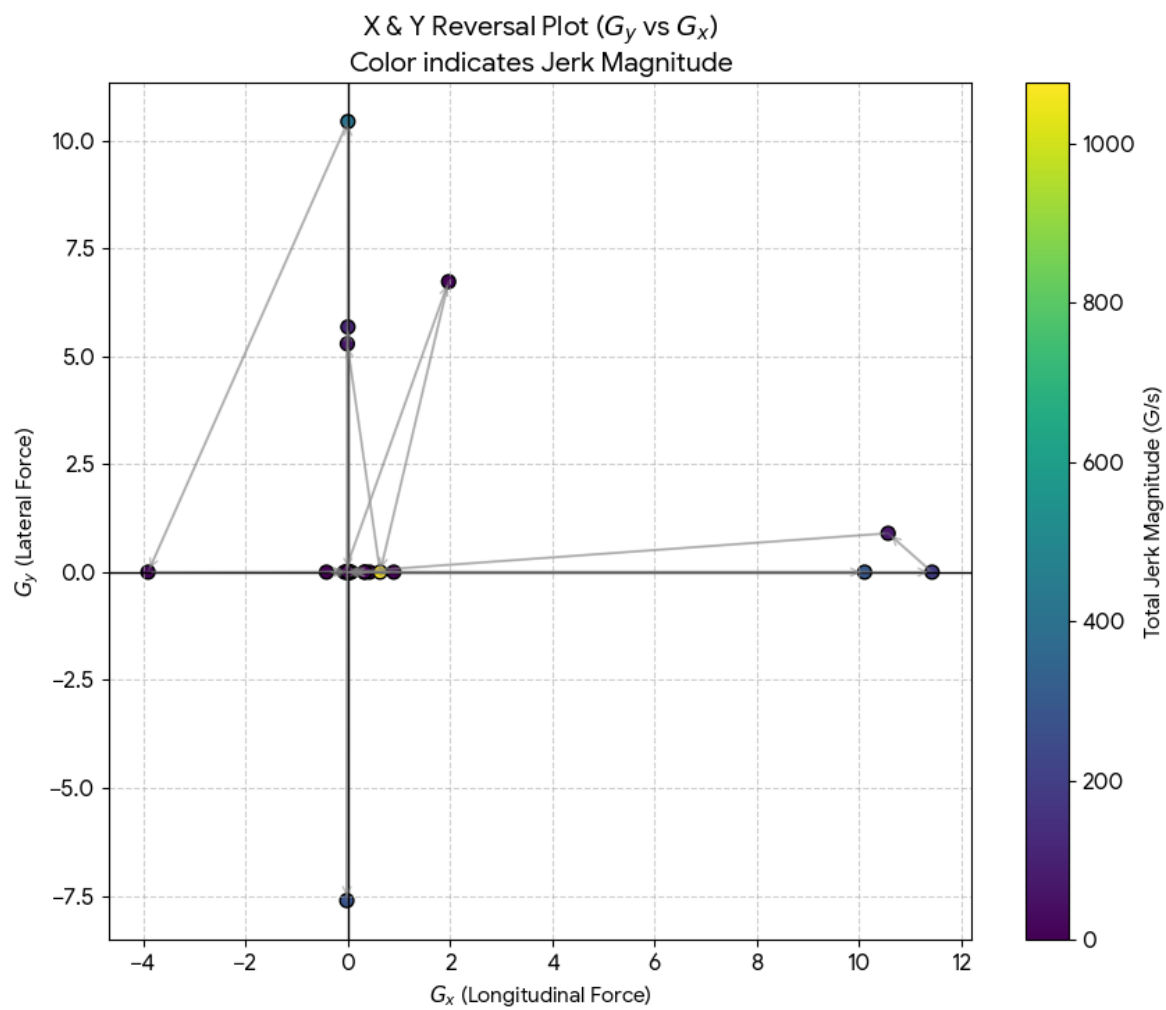


Figure 30: X&Y Reversal Plot (G_y vs G_x)

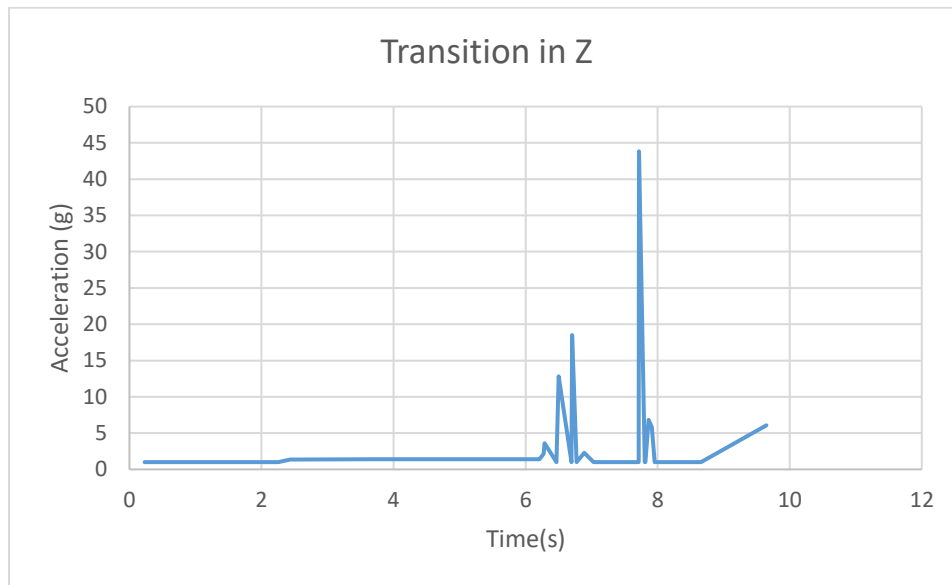


Figure 31: Z-Transitions graph

Joop Adventures Onset calculations

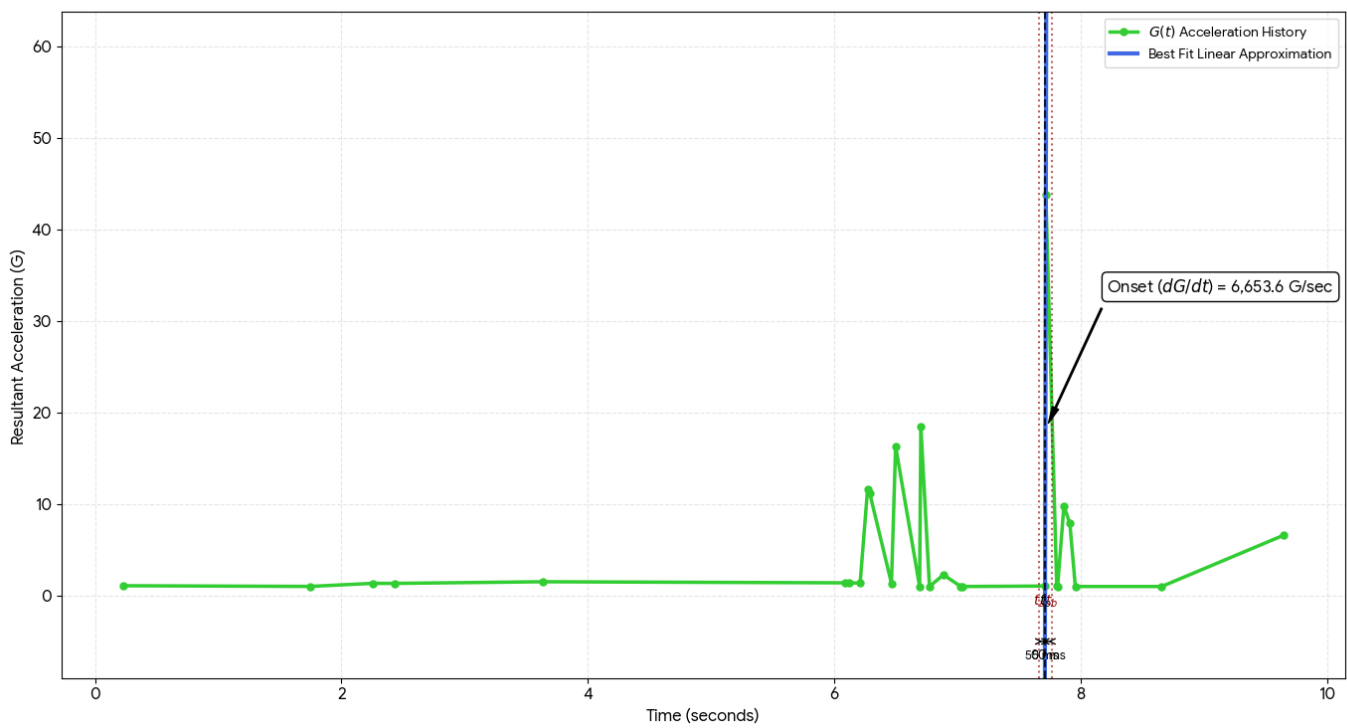


Figure 32: Onset Calculations

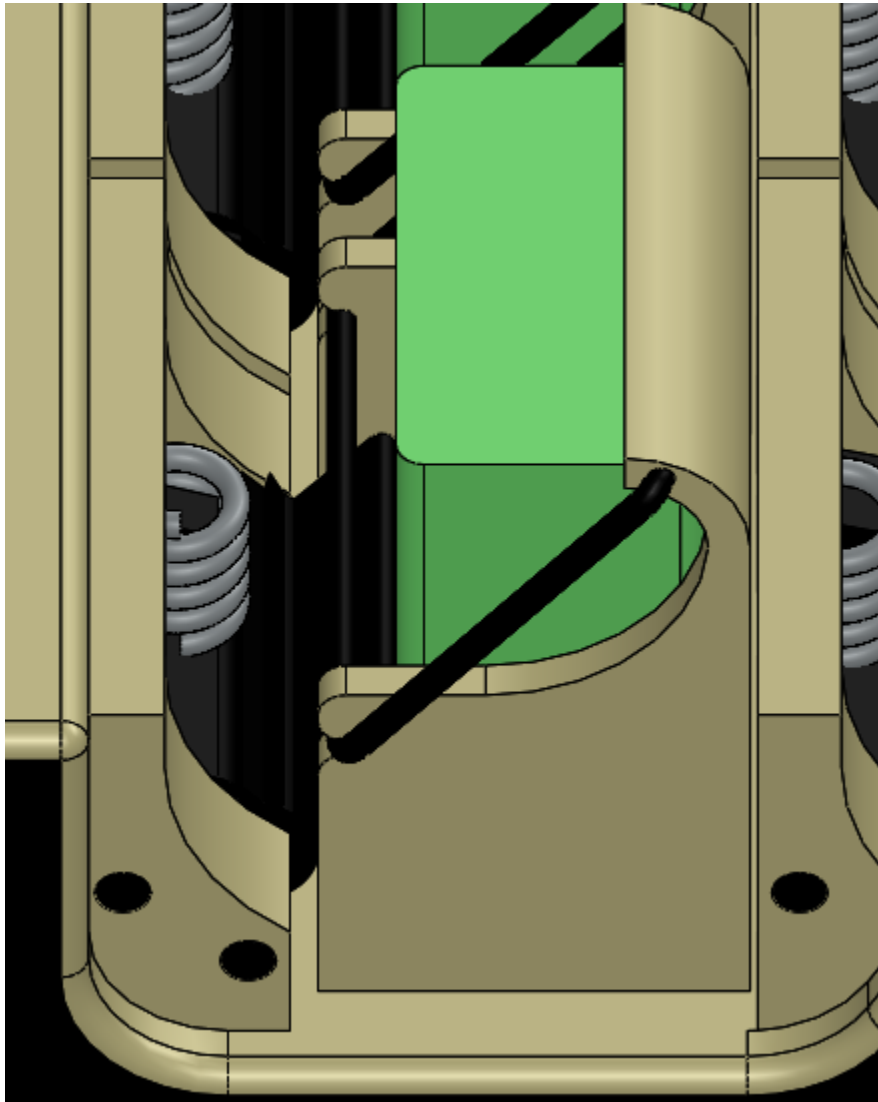


Figure 3326: Lap Bar Restraint System View 1

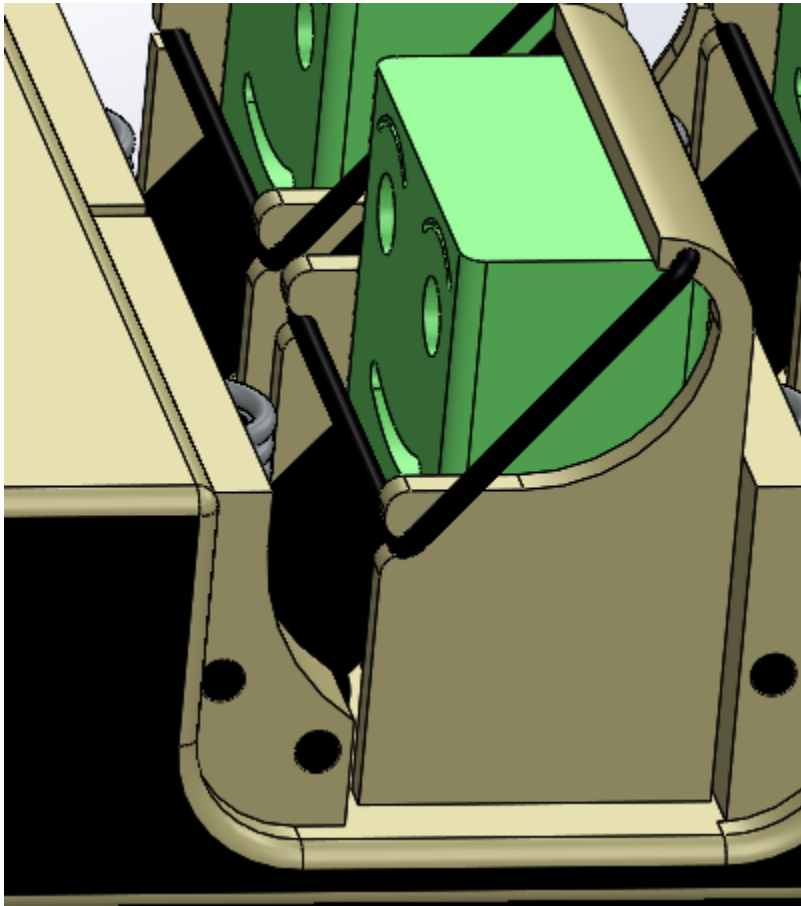


Figure 34: Lap Bar Restraint System View 2

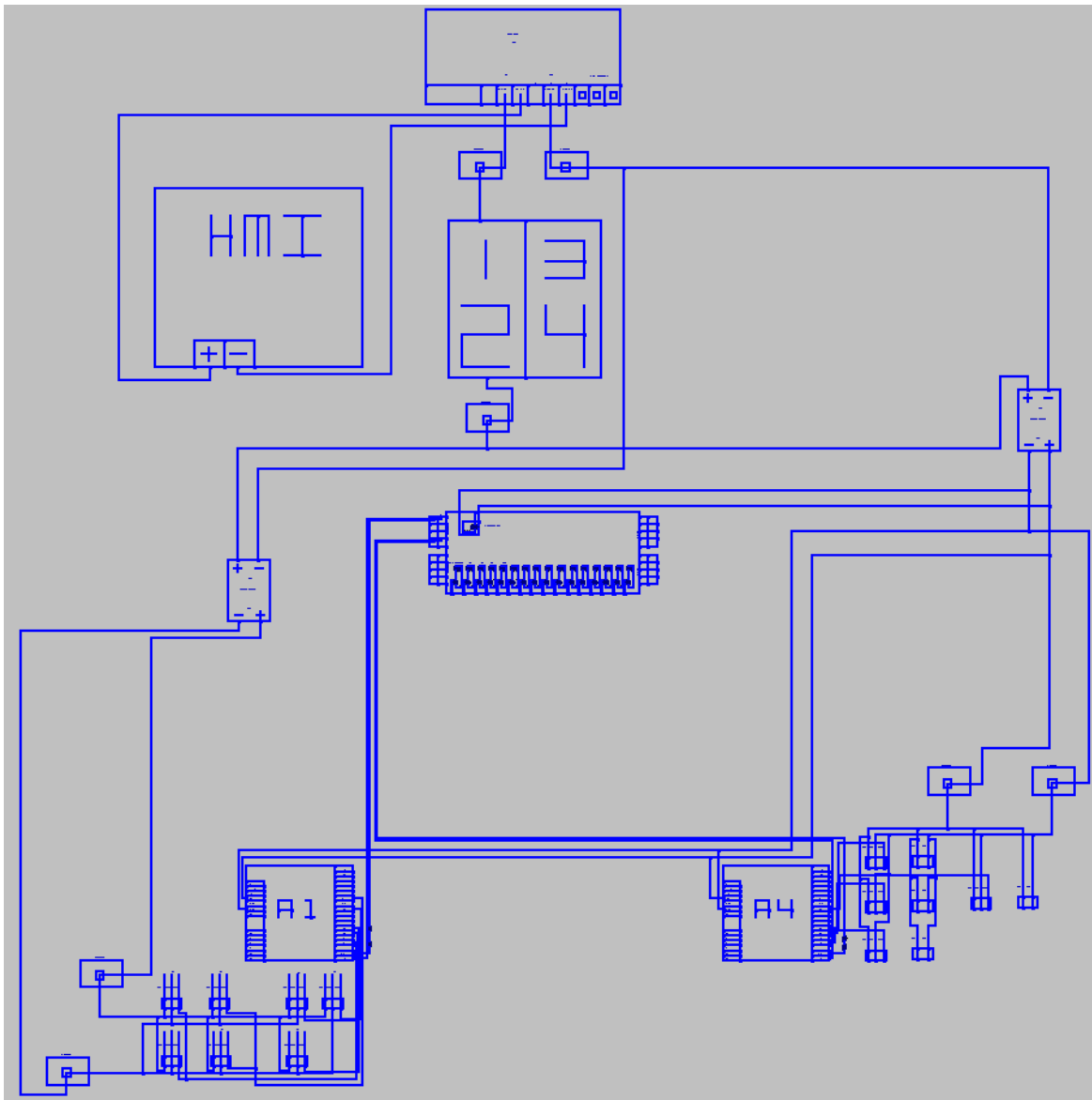


Figure 35: Control System Schematic